

ISA CERTIFIED PERSONAL TRAINER COURSE

WEEK 10

- CHAPTER 13
CONSIDERATIONS FOR CLIENTS
WITH CHRONIC DISEASE
- CHAPTER 15
CONSIDERATIONS FOR CLIENTS WITH
MUSCULOSKELETAL ISSUES



TABLE OF CONTENT



CVD & Cerebrovascular Disorders



Metabolic Disorders



Pulmonary Disorders



Other Chronic Conditions



Musculoskeletal Conditions

CHAPTER 13
CONSIDERATIONS FOR CLIENTS
WITH CHRONIC DISEASE

CHRONIC DISEASES

6 IN 10
Adults in the US
have a chronic disease



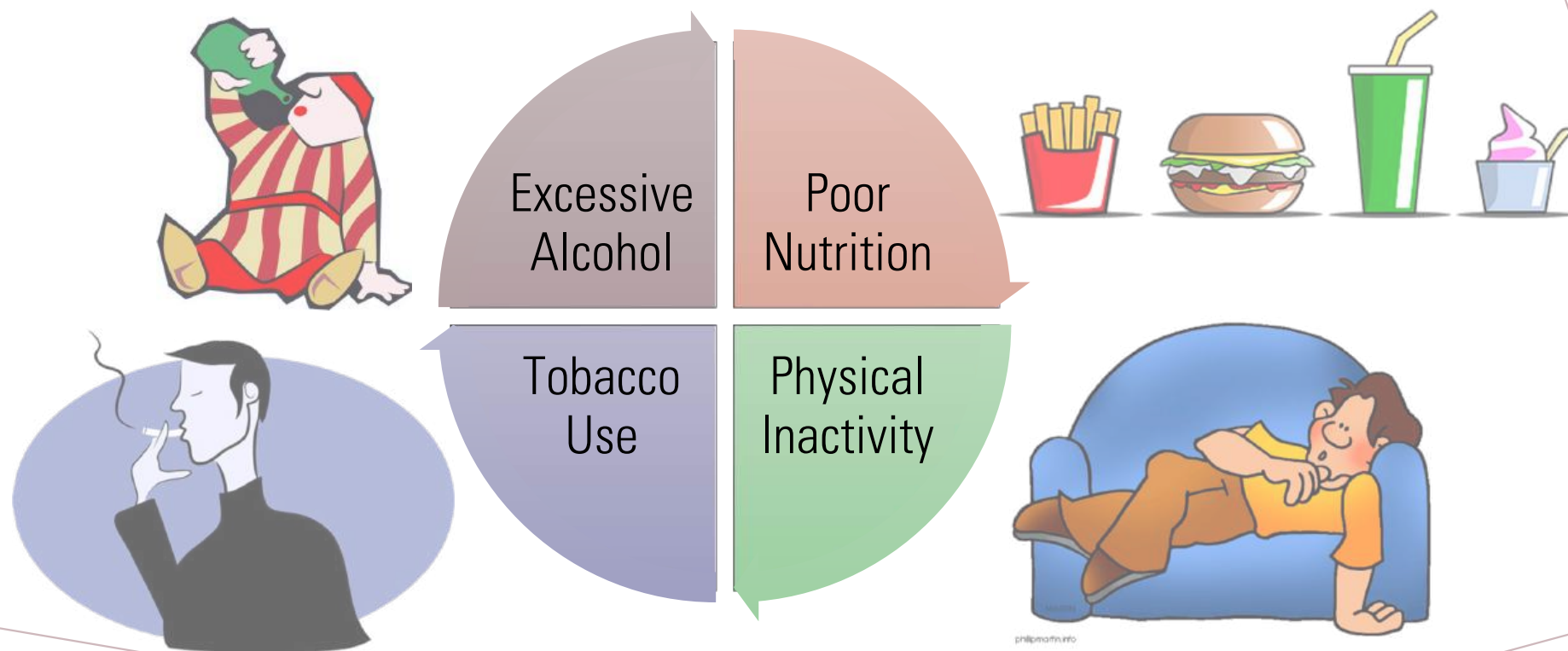
4 IN 10
Adults in the US
have two or more

THE LEADING CAUSES OF DEATH AND DISABILITY



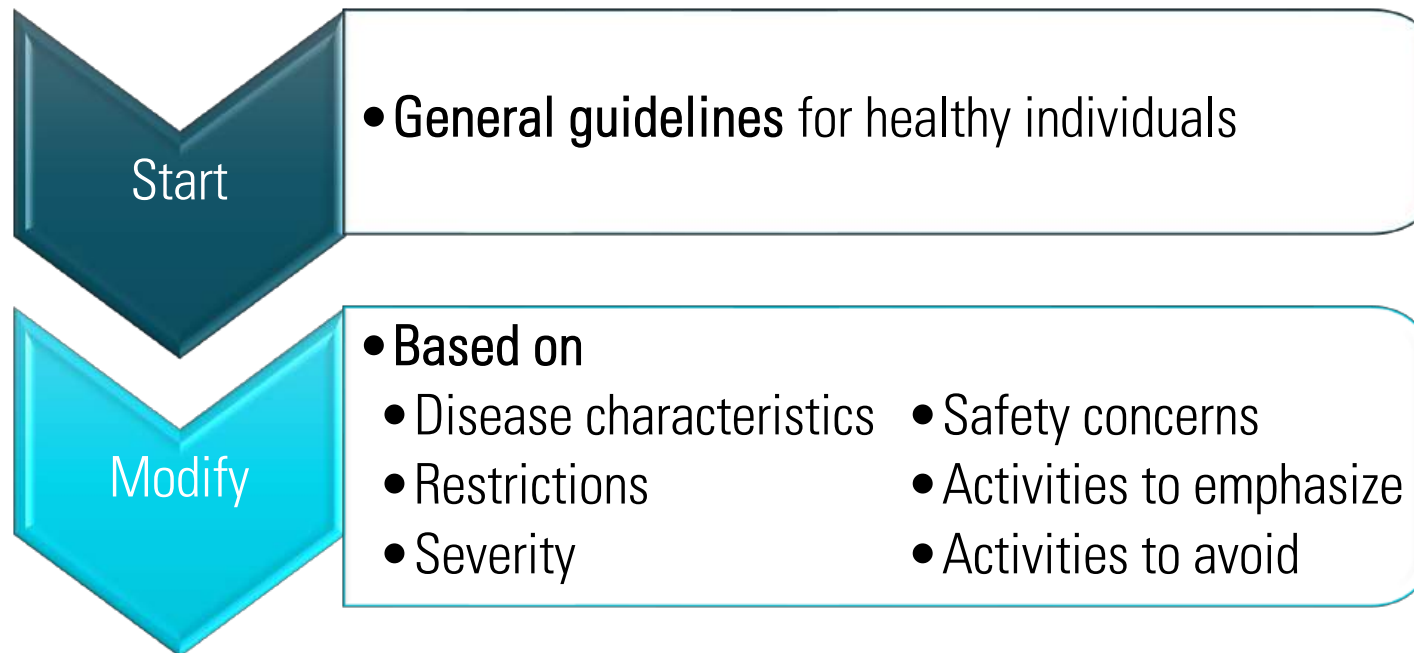
Source from: CDC (Centers for Disease Control and Prevention), "Chronic diseases in America", 2019, Accessed Sep. 4, 2020.

CHRONIC DISEASES AND LIFESTYLE RISK FACTORS



Source from: <https://ephtracking.cdc.gov/showLifestyleRiskFactorsMain>

GENERAL PROGRAM DESIGN PRINCIPLES



Safe
&
Effective

CHRONIC DISEASE

Additional Reading Materials

1. CDC (Centers for Disease Control and Prevention), "Chronic diseases in America", 2019, Accessed Sep. 4, 2020.
<https://www.cdc.gov/chronicdisease/resources/infographic/chronic-diseases.htm>
2. Health Hub, "Chronic Diseases: Understanding the Medical Conditions and their Causes", 2020. https://www.healthhub.sg/a-z/diseases-and-conditions/96/topics_chronic_diseases
3. CDC (Centers for Disease Control and Prevention), "Adult Obesity Facts". 2020.
<https://www.cdc.gov/obesity/data/adult.html>

CARDIOVASCULAR AND CEREBROVASCULAR DISORDERS

- Coronary Artery Disease
- Hypertension
- Stroke
- Peripheral Arterial Disease

HEART DISEASE

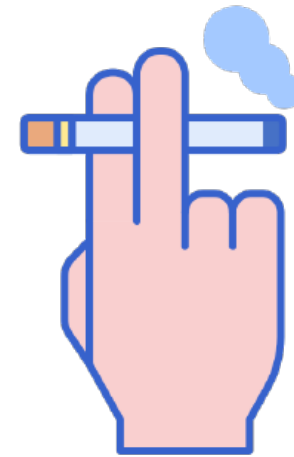
About half of all Americans have at least 1 of the 3 key risk factors:



High Blood Pressure



High Cholesterol

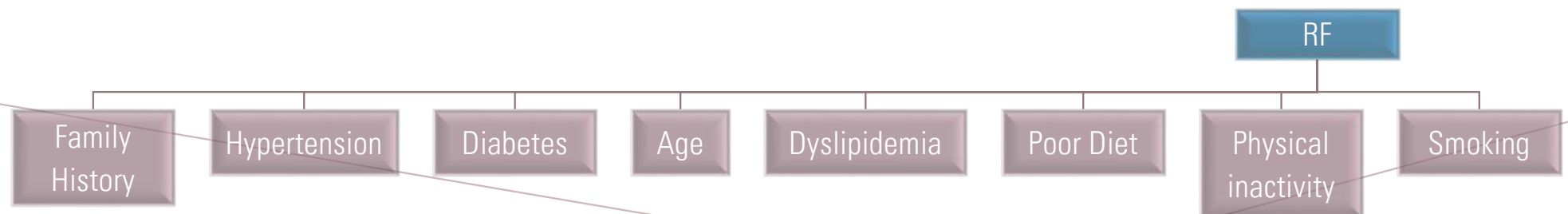
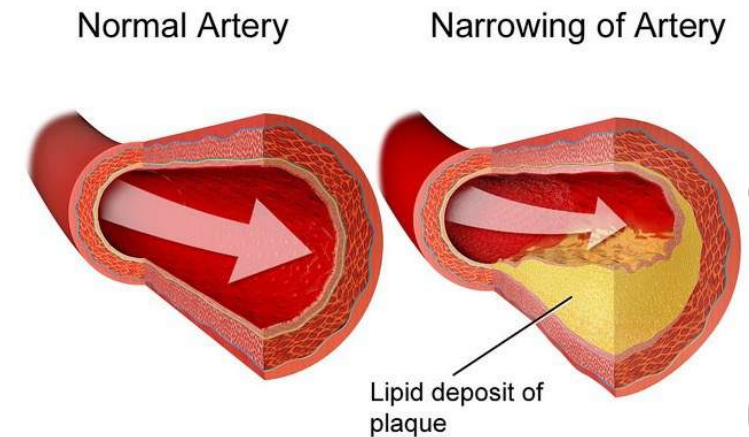
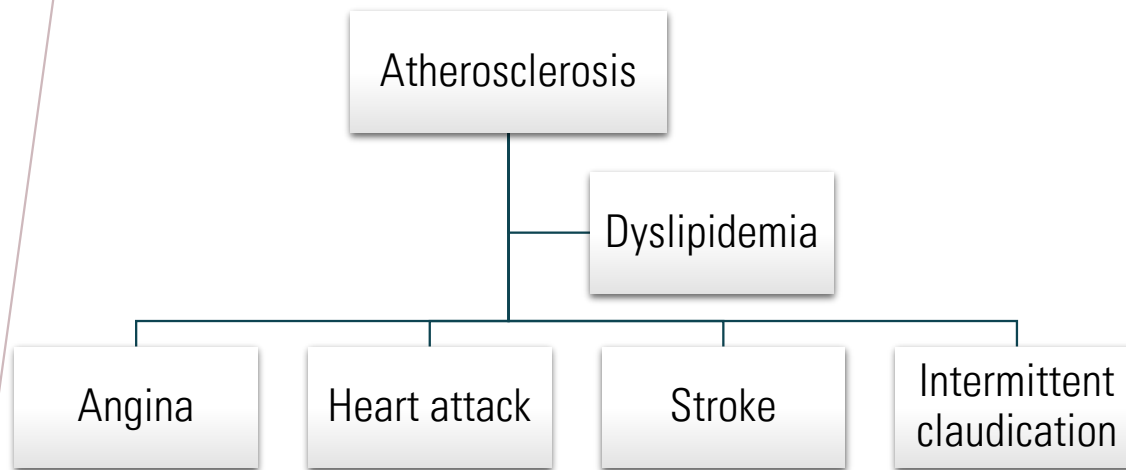


Smoking

CORONARY ARTERY DISEASE (CAD)

[PG 596-597]

Narrowing of the coronary arteries that supply the heart muscle with blood and oxygen



CAD – NUTRITION

Primary

Fruits & veggies ≥ 4.5 cups/d

Fish & shellfish ≥ 2 3.5-oz servings/wk

Sodium ≤ 1500 mg/d

Sugar-sweetened beverage ≤ 36 fl oz/wk

Whole grains ≥ 3 1-oz servings/d

Secondary

Nuts & seeds ≥ 4 1-oz servings/wk

Legumes ≥ 4 ½-cup serving/wk

Processed meats ≤ 2 1.75-oz servings/wk

Sat. fat $\leq 7\%$ of total daily energy intake

CAD – EXERCISE GUIDELINES

	Frequency	Intensity	Time	Type	Progression
CT	3 to ≥ 5 d/wk	Mod.-vig. 40-80%HRR RPE12-16 or RHR+ 20 to 30 HR<10 bpm below ischemic threshold Low-risk, stable CAD clients with base level fitness may perform HIIT	Low fitness clients can start with 1-10 min sessions Eventual goal: 20-60 min	Multiple: e.g. recumbent stepper, elliptical, treadmill, walking, swimming, stationary cycling, or other ergometer use. Continuous or intermittent	Follow ACE-IFT Model
RT	2-3 d/wk	40-60% 1RM or 10-15 reps RPE11-13	8-10 exrs 1 set $\Rightarrow$ 3 sets	All major muscle groups Machine & FWs	Follow ACE-IFT Model

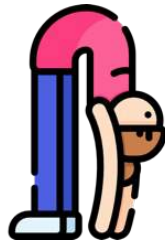
CAD – SAFETY



Obtain physician clearance,
recommendations, and limitations



Proper breathing technique
(avoid Valsalva maneuver)



Proper technique for full,
pain-free ROM



Observe abnormal signs and symptoms
that warrant termination of the exercise

CAD – ADDITIONAL READING MATERIALS

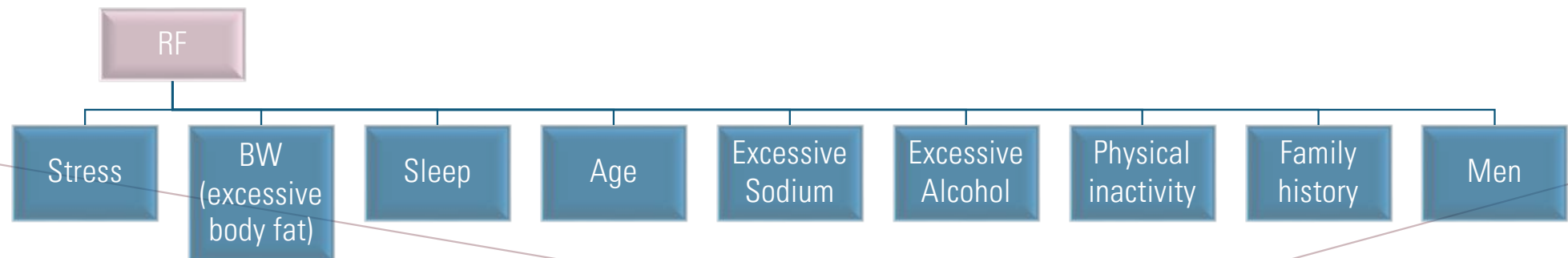
Extra Reading Materials:

1. Centers for Disease Control and Prevention, “Heart Disease”,
<https://www.cdc.gov/heartdisease/index.htm>

HYPERTENSION

Category	Systolic Blood Pressure		Diastolic Blood Pressure
Normal	<120 mmHg	and	<80 mmHg
Elevated	120–129 mmHg	and	<80 mmHg
Hypertension			
Stage 1	130–139 mmHg	or	80–89 mmHg
Stage 2	≥140 mmHg	or	≥90 mmHg

Table 13-2: Categories of Blood Pressure in Adults



HYPERTENSION - TREATMENT

[PG 602-603]

Modification	Recommendation	Approximate SBP Reduction (Range)
Weight reduction	Maintain normal body weight (body mass index 18.5–24.9 kg/m ²)	5–20 mmHg/10 kg weight loss
Adopt DASH eating plan	Consume a diet rich in fruits, vegetables, and low-fat dairy products with a reduced content of saturated and total fat	8–14 mmHg
Dietary sodium reduction	Reduce dietary sodium intake to no more than 100 mmol per day (2.4 g sodium or 6 g sodium chloride)	2–8 mmHg
Physical activity	Engage in regular aerobic physical activity such as brisk walking (at least 30 minutes per day, most days of the week)	4–9 mmHg
Moderation of alcohol	Limit consumption to no more than two drinks per day for most men and to no more than one drink per day for women and lighter-weight persons ^a	2–4 mmHg

Table 13-3: Lifestyle Modifications to Manage Hypertension

HYPERTENSION – EXERCISE GUIDELINES

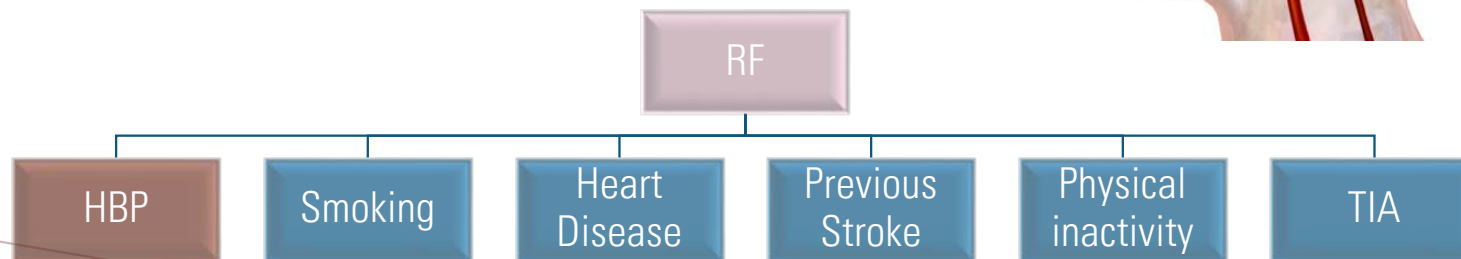
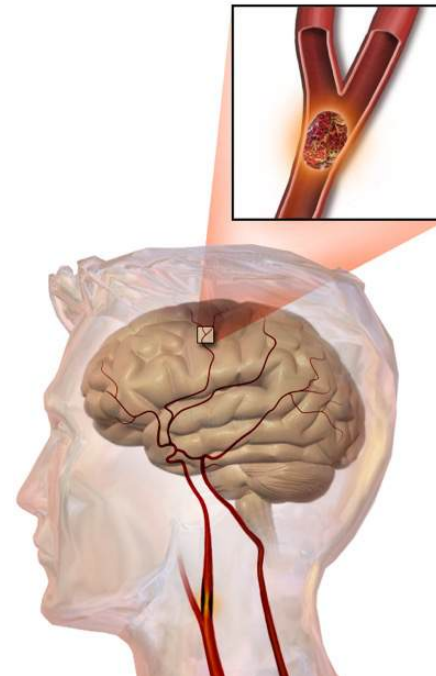
- Obtain the physician's release, guidelines, and recommendations
- Be aware of the client's medications and their potential impact on exercise; use RPE
- Avoid rapid change of position; prolong cool-down
- Maintain hydration status for those on diuretics
- Avoid isometrics and inverted exercises; emphasize technique and breathing
- Measure the client's pre- and post-exercise blood pressure:
 - Discontinue exercise if: SBP>250 mmHg or DBP>115 mmHg, or SBP fails to increase/drops by more than 20 mmHg with increasing workload

HYPERTENSION – EXERCISE GUIDELINES

	Frequency	Intensity	Time	Type	Progression
CT	≥ 5 d/wk Preferably 7 d/wk	40-60%HRR, RPE12-13 or HR below VT1	≥ 30 min continuous or cumulative	E.g. walking, jogging, cycling, swimming, elliptical training, stairclimbing	Follow ACE-IFT Model Consider medication effects, drug changes, and comorbidities
RT	2-3 d/wk	60-80% 1RM; 40-50% 1RM for older adults/novice	2-4x8-12	All major muscle groups Machine & FWs	Follow ACE-IFT Model

STROKE

- Ischemic stroke
Blockage in a vessel in the brain
- Hemorrhagic stroke
Blood vessel in the brain ruptures



STROKE – SIGNS & SYMPTOMS

To spot a Stroke
think F.A.S.T.

**SPOT
STROKE**
Fast discovery helps recovery



Face drooping
Is the person's
smile uneven?



Arm weakness
Can the person raise both
arms and keep them up?



Speech difficulty
Does the person's speech
sound slurred or unclear?



Time to call 995
If the person shows any of these signs, they
need to be rushed to the hospital immediately.

STROKE – EXERCISE GUIDELINES

Considerations:

- Disability
- Physical deconditioning
- Muscular atrophy
- Increased CVD risk, recurring stroke and MI

Training Focus:

- Optimize basic ACL skills
- Regain balance, coordination, and functional independence
- Prevent complications and stroke reoccurrence
- Use a variety of exercises to improve functional capacity, balance, and coordination
- Modify according to individual client's profile

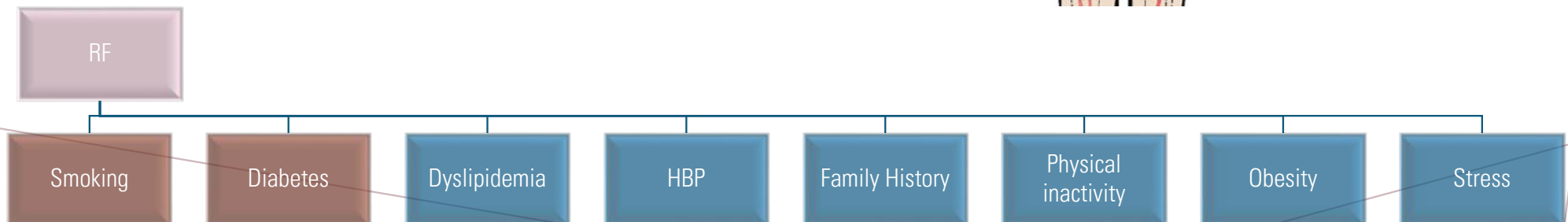
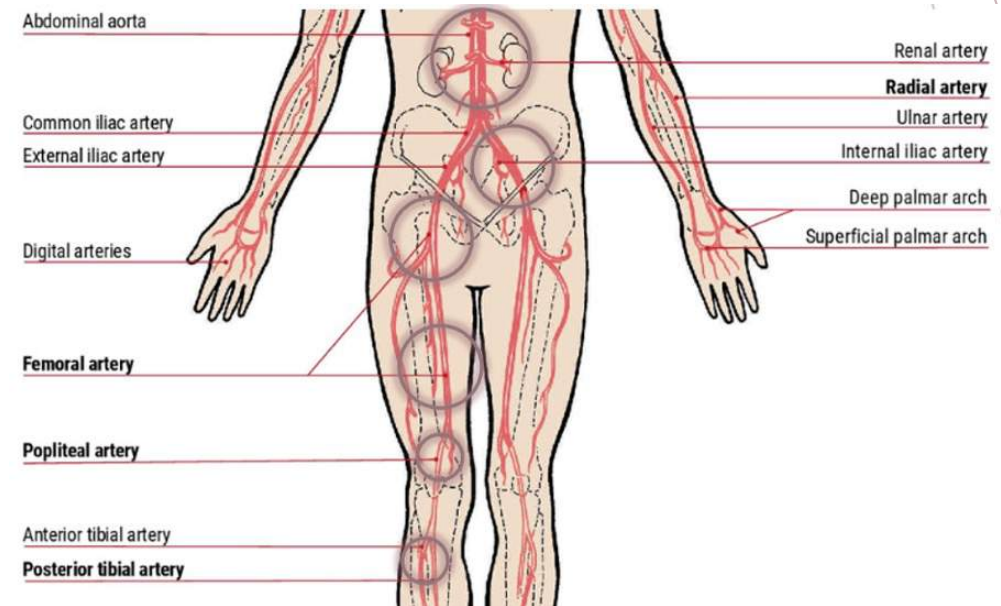


STROKE – EXERCISE GUIDELINES

	Frequency	Intensity	Time	Type	Progression
CT	3-5 d/wk	Light, moderate, or vigorous 40-70%HRR (with GXT data) RPE11-14	20-60 min Consider multiple bouts throughout the day	Consider func'l & cognitive deficiencies; some equipment may need modifications. • Cycle ergometry • Recumbent stepper • Treadmill walking	Follow ACE-IFT Model
RT	2 d/wk	50-70% 1RM	1-3x8-15	All major muscle groups Equipment and higher safety exrs	Follow ACE-IFT Model

PERIPHERAL ARTERIAL DISEASE (PAD)

- PAD: Muscular pain caused by ischemia (lack of blood flow) to the muscle
- Often a result of atherosclerosis
- Claudication: cramping leg pain induced by exercise



PAD – NUTRITION

Primary

Fruits & veggies ≥ 4.5 cups/d

Fish & shellfish ≥ 2 3.5-oz servings/wk

Sodium ≤ 1500 mg/d

Sugar-sweetened beverage ≤ 36 fl oz/wk

Whole grains ≥ 3 1-oz servings/d

Secondary

Nuts & seeds ≥ 4 1-oz servings/wk

Legumes ≥ 4 ½-cup serving/wk

Processed meats ≤ 2 1.75-oz servings/wk

Sat. fat $\leq 7\%$ of total daily energy intake

PAD - EXERCISE GUIDELINES

- Obtain medical clearance and exercise guidelines from client's healthcare providers
- Walk to a pain scale of 3-4. Rest. Continue to walk. Repeat
- Do not exceed a pain scale of 4.
- Use the pain scale instead of %VO₂max to measure exercise intensity
- Watch for CVD symptoms
- Ensure proper footwear and pay close attention to client's feet

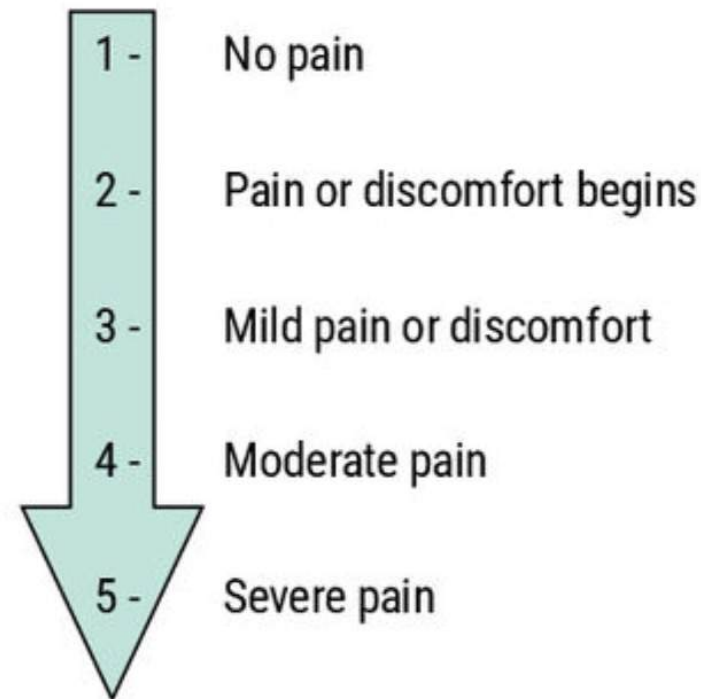


Figure 13.1: Claudication pain / discomfort scale

PAD - EXERCISE GUIDELINES

	Frequency	Intensity	Time	Type	Progression
CT	3-5 d/wk	Moderate <VT1 @a pace/grade that triggers claudication pain in 2-6 min. Do not exceed pain scale 4	30-45 min (excl. rest periods) for up to 12 wks ⇒ 60 min	Weight-bearing. • Walking • Intermittent exr with seated rest @ pain scale 4; resume when pain dissipates	Follow ACE-IFT Model
RT	2 d/wk	60-80% 1RM	6-8 exrs 2-3x8-12	Whole body, focus on large muscle groups Emphasize lower-limb exrs	Follow ACE-IFT Model

Table 13-6: Exercise Guidelines Summary for Clients with Peripheral Arterial Disease

PAD – ADDITIONAL READING MATERIALS

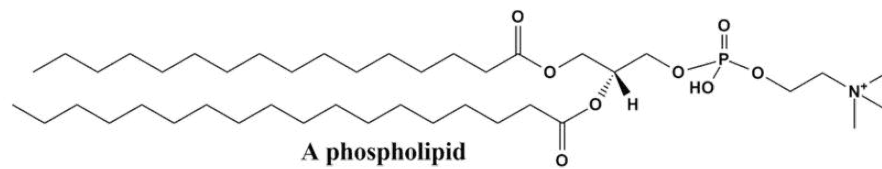
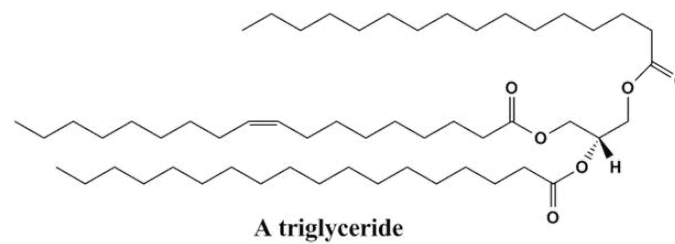
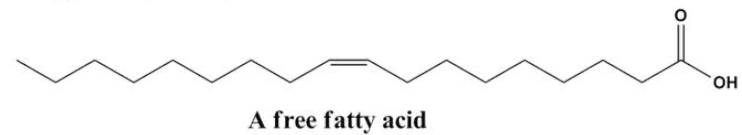
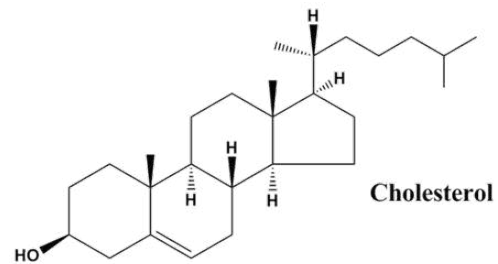
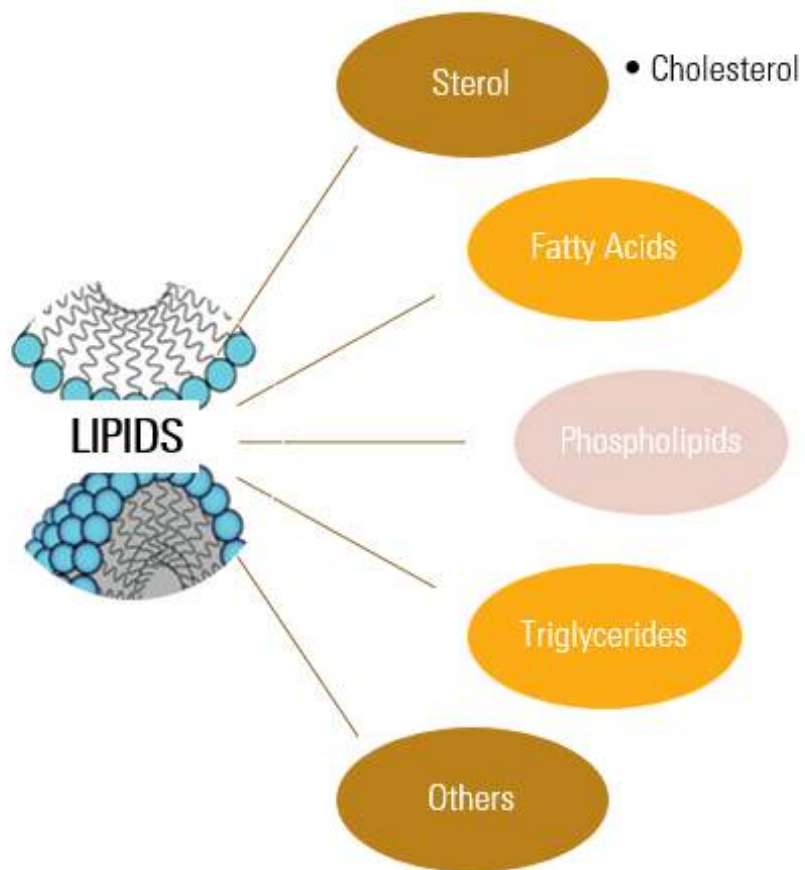
Additional Reading Materials

1. N. M. Hamburg and G. J. Balady, "Exercise Rehabilitation in Peripheral Artery Disease: Functional Impact and Mechanisms of Benefits," *Circulation*, vol. 123, no. 1, pp. 87–97, Jan. 2011, doi: [10.1161/CIRCULATIONAHA.109.881888](https://doi.org/10.1161/CIRCULATIONAHA.109.881888).
2. A. T. Hirsch *et al.*, "ACC/AHA 2005 Practice Guidelines for the Management of Patients With Peripheral Arterial Disease (Lower Extremity, Renal, Mesenteric, and Abdominal Aortic). *Circulation*, vol. 113, no. 11, Mar. 2006, doi: [10.1161/CIRCULATIONAHA.106.174526](https://doi.org/10.1161/CIRCULATIONAHA.106.174526).

METABOLIC DISORDERS

- Dyslipidemia
- Diabetes
- Metabolic Syndrome

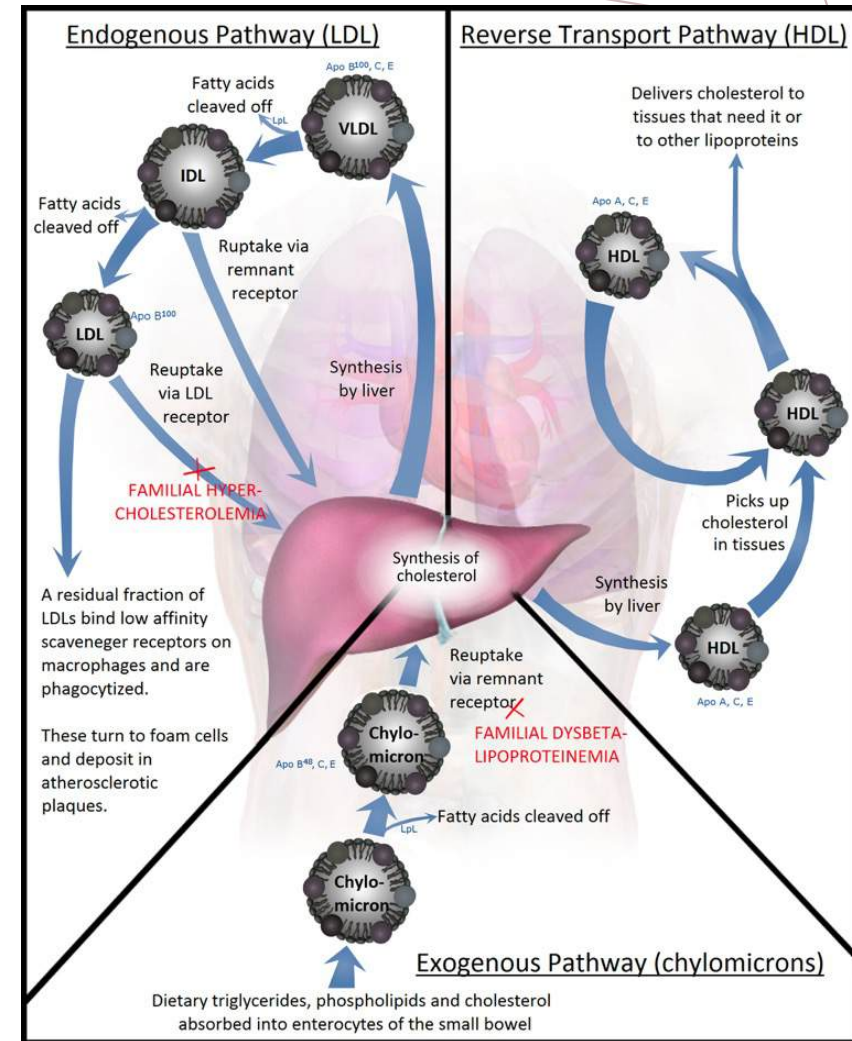
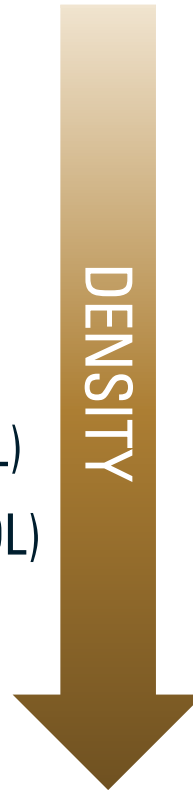
LIPIDS



LIPOPROTEINS



- Chylomicrons (ULDL)
- Very low-density lipoprotein (VLDL)
- Intermediate-density lipoprotein (IDL)
- Low-density lipoprotein (LDL)
- High-density lipoprotein (HDL)



Source from: Nicholas Patchett

DYSLIPIDEMIA

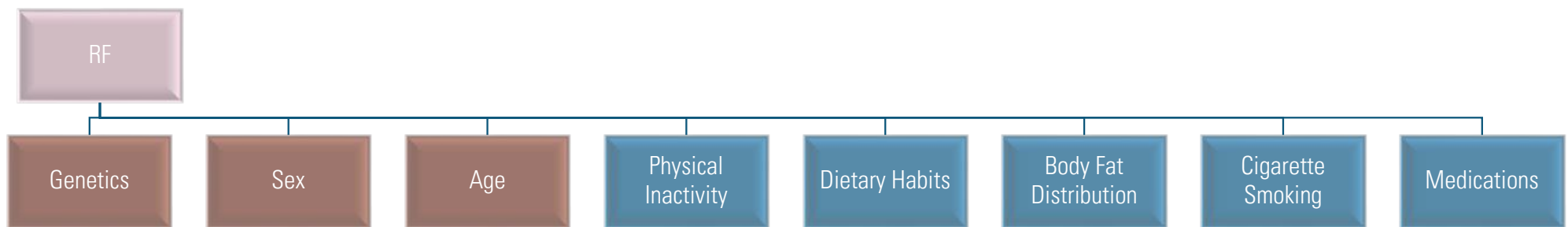
LDL Cholesterol		HDL Cholesterol	
Optimal*	<100	Low	<40
Near optimal/above optimal	100–129	High	≥60
Borderline high	130–159	Total Cholesterol	
High	160–189	Desirable	<200
Very high	≥190	Borderline high	200–239
		High	≥240

Table 13-7: ATP III Classification of LDL, Total Cholesterol, and HDL Cholesterol (mg/dL)

Normal	<150
Borderline high	150–199
High	200–499
Very high	≥500

Table 13-8: Classification of Triglycerides (mg/dL) (Fasting)

DYSLIPIDEMIA – RISK FACTORS



DYSLIPIDEMIA – NUTRITION

- Emphasize vegetables, fruits, and whole grains
- Include low-fat dairy products, poultry, fish, legumes, non-tropical vegetable oils, and nuts
- Limit sweets, sugar-sweetened beverages, red meats
- Limit calories from saturated fat (aim for 5-6% of calories)
- Reduce or eliminate trans fat consumption



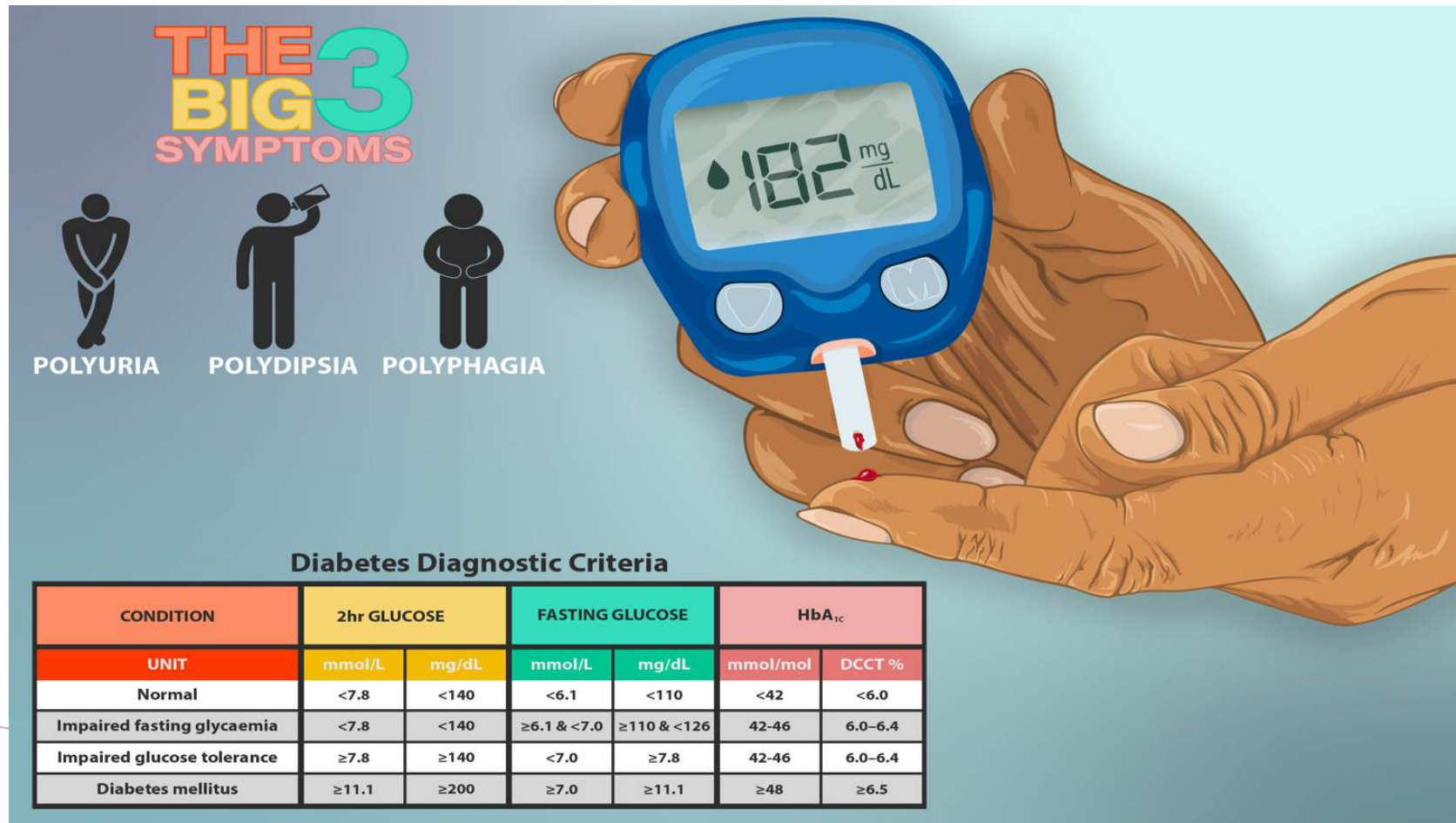
DYSLIPIDEMIA – EXERCISE GUIDELINES

	Frequency	Intensity	Time	Type	Progression
CT	≥5 d/wk	Moderate to vigorous From <VT1 to <VT2 40-75% HRR	30-60 min per session ≥ 50-60 min/d	Varies	Follow ACE-IFT Model
RT	2-3 d/wk	50-85% 1RM (strength) <50% 1RM (endurance)	2-4x8-12 (S) or 2-4x12-20 (E)	All major muscle groups Machines, FW, BW	Follow ACE-IFT Model

Table 13-9: Exercise Guidelines Summary for Clients with Dyslipidemia

DIABETES

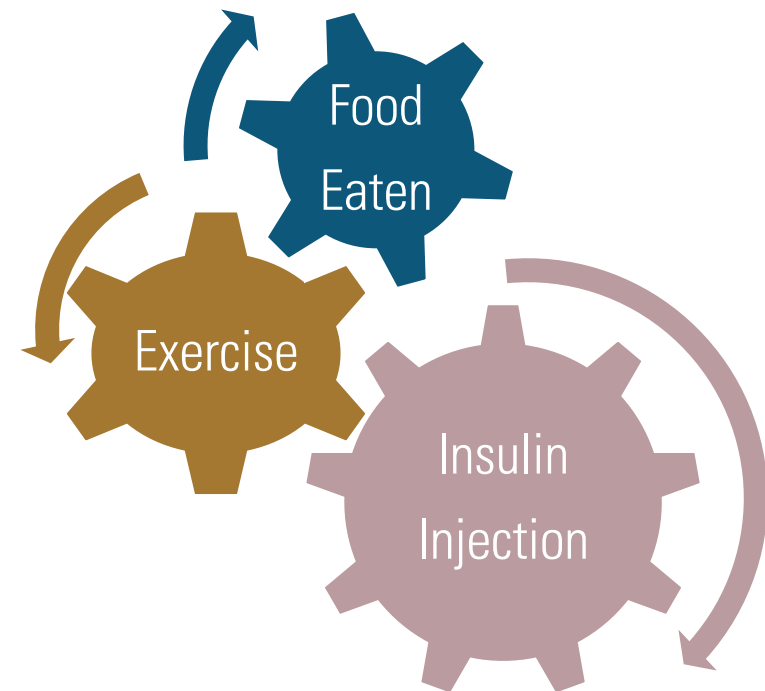
[PG 614-615]



DIABETES – TYPE 1

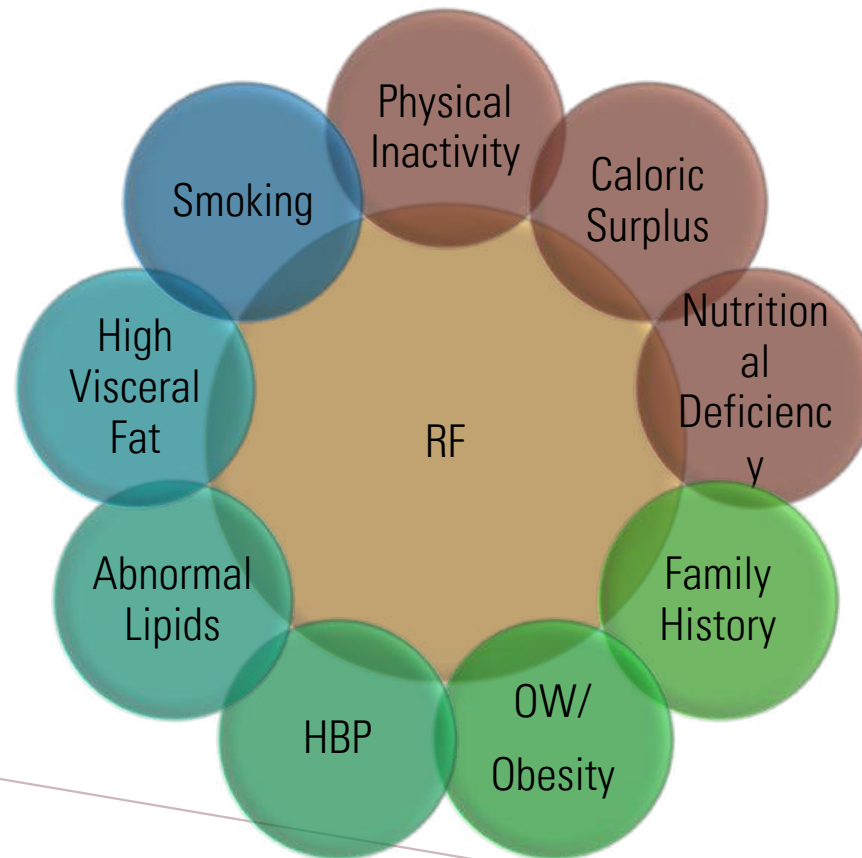
Autoimmune disease – beta cells are destroyed

- Symptoms of hyperglycemia appear
- Consult with client's healthcare provider
- Follow general exercise guidelines
- Reduce insulin dose before exercise, take fast-acting CHO before and during exercise, adjust infusion from the insulin pump during and after exercise



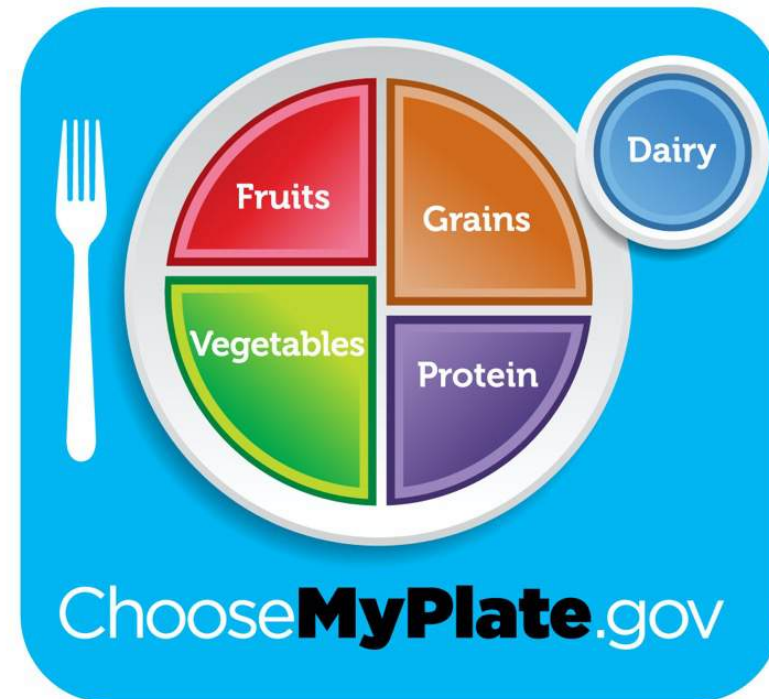
DIABETES – TYPE 2

Insulin resistance / insensitivity



DIABETES – TYPE 2 – NUTRITION

- Dietary Guidelines for Americans
- MyPlate Guidelines for Singaporeans
- Consume complex carbohydrates instead of simple carbohydrates
- Monitor blood glucose level before and 2h after meals



DIABETES – EXERCISE GUIDELINES

	Frequency	Intensity		Time	Type	Progression
CT	3-7 d/wk	Moderate <VT1 40-60% HRR RPE 12-13	Vigorous VT1 to <VT2 60-90% HRR RPE 14-17	Type 1: Same as general pop Type 2: 150 min/wk mod- intensity	Varies	Follow ACE- IFT Model
RT	≥2 d/wk	50-85% 1RM		8-10 exr to near fatigue 1-3x8-10 (10-15 reps to start)	All major muscle groups Machines, FW	Follow ACE- IFT Model

DIABETES – SAFETY

- Check blood sugar before, right after, and in the hours following an exercise
- Check blood sugar every 30 to 60 minutes during prolonged workout
- Start the exercise with blood glucose ≥ 100 mg/dL (5.6 mmol/L) and ≤ 250 mg/dL (13.9 mmol/L) (unless no ketosis detected)
- If blood glucose is below 100 mg/dL, administer 15-20 g of fast-acting CHO



DIABETES – SAFETY

- Hypoglycemia symptoms:
 - Shaking, nervous/anxious, sweating, chills and clamminess, irritability/impatience, fast heartbeat, hunger, nausea, pallor, feeling sleepy, weak, or having no energy, tingling or numbness in the lips, tongue or cheeks, nightmares or crying out during sleep
 - Confusion, blurred vision, difficulty concentrating, slurred speech, numbness, drowsiness
 - Seizures, fainting/coma
 - Death
- 15-15 Rule
 - Glucose tablet
 - Gel tube
 - ½ cup juice or regular soda
 - 1 tbs sugar, honey or corn syrup
 - Hard candies, jellybeans or gumdrops

DIABETES – ADDITIONAL READING MATERIALS

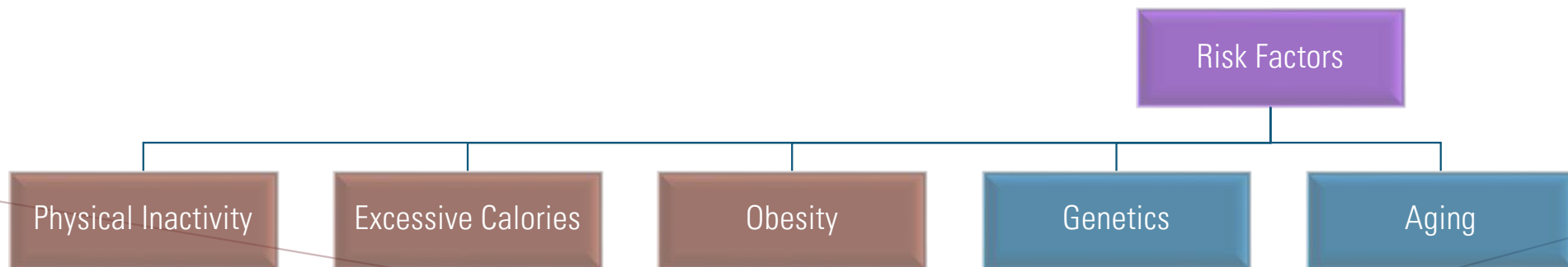
Extra Reading Materials

1. WHO, “Diabetes Action Now: An initiative of the World Health Organization and the International Diabetes Federation”, 2004.
<https://www.who.int/diabetes/actionnow/en/DANbooklet.pdf>
2. WHO Fact Sheets, “Diabetes”, 2020. <https://www.who.int/news-room/fact-sheets/detail/diabetes>
3. American Diabetes Association, “Blood Sugar and Exercise”.
<https://www.diabetes.org/fitness/get-and-stay-fit/getting-started-safely/blood-glucose-and-exercise>

METABOLIC SYNDROMES (METS)

- Having ≥ 3 of the following conditions:

Increased BP	Elevated fasting blood glucose	Elevated waist circumference	Reduced HDL cholesterol	Elevated TG
• $\geq 130/85$ mmHg or on drug treatment	• ≥ 100 mg/dL or on drug treatment	• $\geq 40/35$ inches or 102/88 cm (men/women)	• $< 40/50$ mg/dL (men/women) or on drug treatment	• ≥ 150 mg/dL or on drug treatment



METS – LIFESTYLE CHANGE TREATMENT



Weight Loss



Physical Activity



Healthy Eating



Tobacco Cessation

METS – EXERCISE GUIDELINES

	Frequency	Intensity	Time	Type	Progression
CT	≥5 d/wk Maximize caloric expenditure	<VT1 40-60% HRR RPE 12-13	30-60 min/session 50-60 min/session preferred Intermittent cumulative or continuous	Varies	Follow ACE-IFT Model
RT	≥2 d/wk	50-85% 1RM	8-10 exr to near fatigue 1-3x8-10	All major muscle groups Machines, FW	Follow ACE-IFT Model

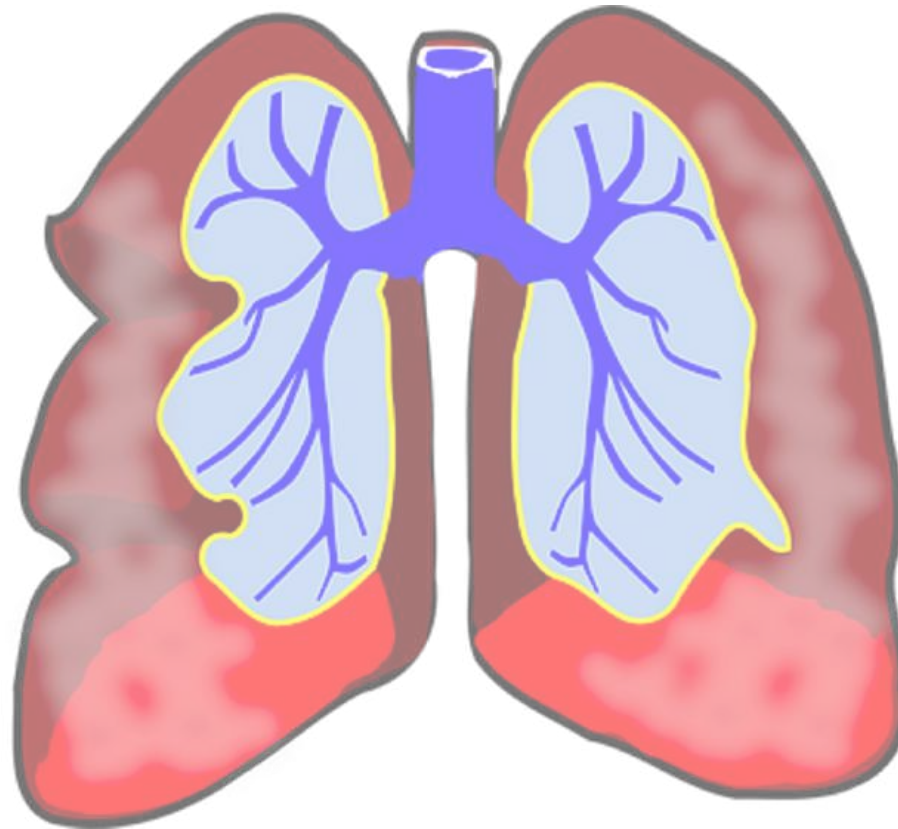
Table 13-12: Exercise Guidelines Summary for Clients with Metabolic Syndrome

METS

- Extra Reading Materials:

1. National Heart, Lung, and Blood Institute (NHLBI), "Metabolic Syndrome", <https://www.nhlbi.nih.gov/health-topics/metabolic-syndrome>

PULMONARY DISORDERS



- COPD
- Asthma

CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)



Source from: https://commons.wikimedia.org/wiki/File:Respiratory_Tract.jpg

ASTHMA

- A chronic inflammatory disorder that causes narrowing of the airways
- Prevalence:
 - Boys 11.1% vs. girls 7.8%
 - Men 5.7% vs. women 9.7%



Asthma Triggers

ASTHMA – LIFESTYLE

- Nutrition influences asthma in several ways:
 - Dietary habits that result in overweight ↗ risk of developing asthma
 - Obesity is associated with severity and frequency of asthma attacks
 - Food allergies are a possible trigger to increase inflammation and asthma attacks
- Smoking
 - Influence the onset and severity of asthma



ASTHMA – EIS

- Exercise-Induced Bronchoconstriction (EIS)
 - An asthmatic attack following moderate to vigorous exercise
 - Severity of EIS is related to the intensity of exercise
 - Symptoms usually peak 5-10 min after exercise and last for 20-30 min
 - Some might develop hacking cough 2-12 h after exercise that can last for 1-2 d
 - ~50% of EIB leads to a relative refractory period (up to 2 h)
 - ~50% of EIB leads to mild asthmatic response 6-8 h after the initial bronchoconstriction
- Strategy to reduce EIS
 - Breathe in through nose to humidify and warm up the air

ASTHMA – EXERCISE GUIDELINES

- Client to review the use of medications with their physician
- Have rescue medication at all times
- Avoid asthma triggers; consider moving indoors on extreme hot or cold days or when pollen counts/air pollution are high. Consider wearing a face mask when exercising in cold
- Plenty of hydration before, during and after exercise
- Prolonged warm-up: 10-15 min vigorous/interval; prolonged cool-down
- Start low and increase intensity slowly, contingent on client's condition
- Closely observe for early signs of EIB
- If an asthma attack is not relieved by medication, activate emergency medical system
- Asthmatics often respond best to exercise in mid-to-late morning

ASTHMA – EXERCISE GUIDELINES [PG 627]

	Frequency	Intensity	Time	Type	Progression
CT	3-5 d/wk	<VT1, 40-60% HRR, RPE 12-13 ↓ (if well tolerated) ↓ (after 1 month) 60-70% HRR	→ 30-40 min/session	E.g. walking, running, cycling, swimming in a nonchlorinated pool	Follow ACE-IFT Model
RT	≥2 d/wk	50-85% 1RM (lower intensity to start)	8-10 exr to near fatigue 1-3x8-10 (10-15 to start)	All major muscle groups Machines, FW	Follow ACE-IFT Model

Table 13-13: Exercise Guidelines Summary for Clients with Asthma

PULMONARY DISORDERS

- Extra Reading Materials

1. NHLBI, "COPD", <https://www.nhlbi.nih.gov/health-topics/copd>
2. American Lung Association, "Learn About Asthma", <https://www.lung.org/lung-health-diseases/lung-disease-lookup/asthma/learn-about-asthma>



OTHER CHRONIC CONDITIONS

- Cancer
- Osteoporosis
- Arthritis

CANCER

- A collection of diseases characterized by uncontrolled spread and growth of abnormal cells



1 in **6** DEATHS
Globally (2018)*

- 87% U.S. cases are above 50-year-old (2017)

LIFETIME RISK OF DEVELOPING CANCER

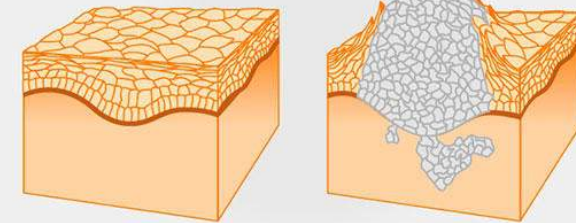
41 out of **100**  **38** out of **100** 

Source: * WHO, "Cancer", 2018. https://www.who.int/health-topics/cancer#tab=tab_1

1 OF 10 | How Does Cancer Form?

NORMAL CELLS

CANCER CELLS



2 OF 10 | What Causes Cancer?

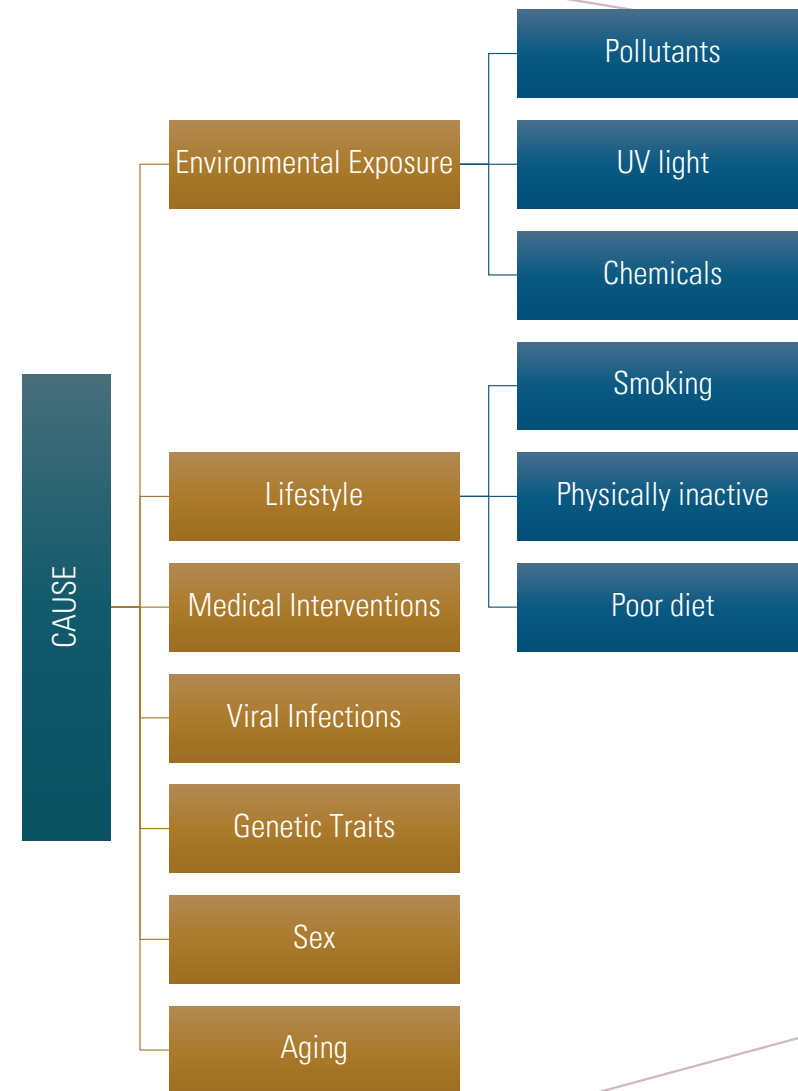
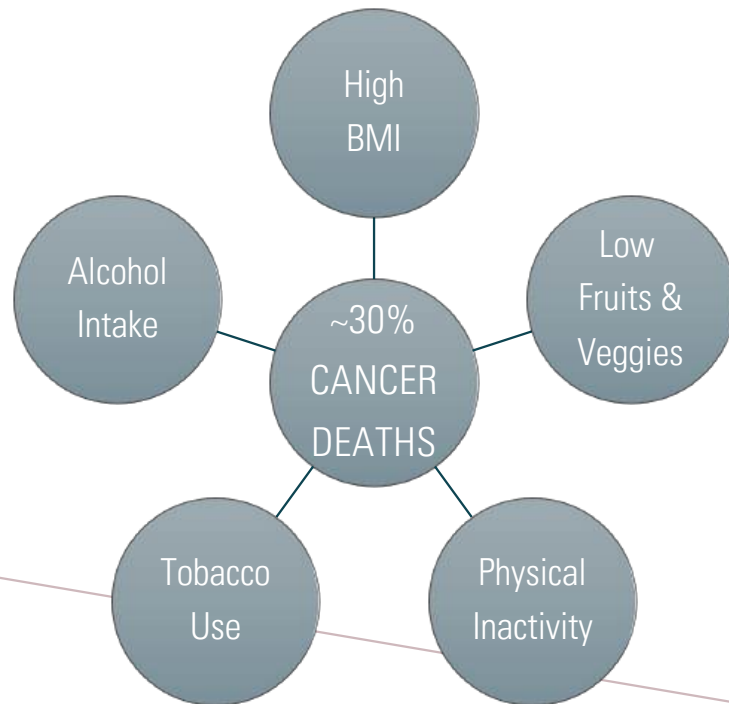
GENE



Source from: <https://www.cancer.gov/about-cancer/understanding/what-is-cancer>

CANCER

30-50% of cancers are preventable.
Tobacco use is the single largest preventable cause of cancer, responsible for ~22% of all cancer-related deaths (2018)



CANCER – NUTRITION



CANCER – EXERCISE

Prevention:

- People with higher physical activity level have reduced likelihood of cancer

Benefits for cancer patients:

- Reduce common side effects of cancer and its treatment: anxiety, depressive symptoms, fatigue, weakened physical functions
- Preserve muscle mass, increase muscular strength and endurance
- Improve balance and overall physical function
- Decrease risk for heart disease, osteoporosis, diabetes



CANCER – EXERCISE GUIDELINES

[PG 632]

	Frequency	Intensity	Time	Type	Progression
CT	3-5 d/wk	<VT1, 40-60% HRR, RPE 12-13 ↓ VT1 to <VT2, 60-90% HRR	75 min/wk vigorous, or 150 min/wk moderate, or combination thereof	Prolonged CR exercise, e.g. walking, cycling	Follow ACE-IFT Model
RT	2-3 d/wk	Start with <30% 1RM, progress slowly (~5%)	≥1x8-10	All major muscle groups Machines, FW Weight-bearing functional tasks Stretching Recreational activities	Follow ACE-IFT Model

Table 13-15: Exercise Guidelines Summary for Clients with Cancer

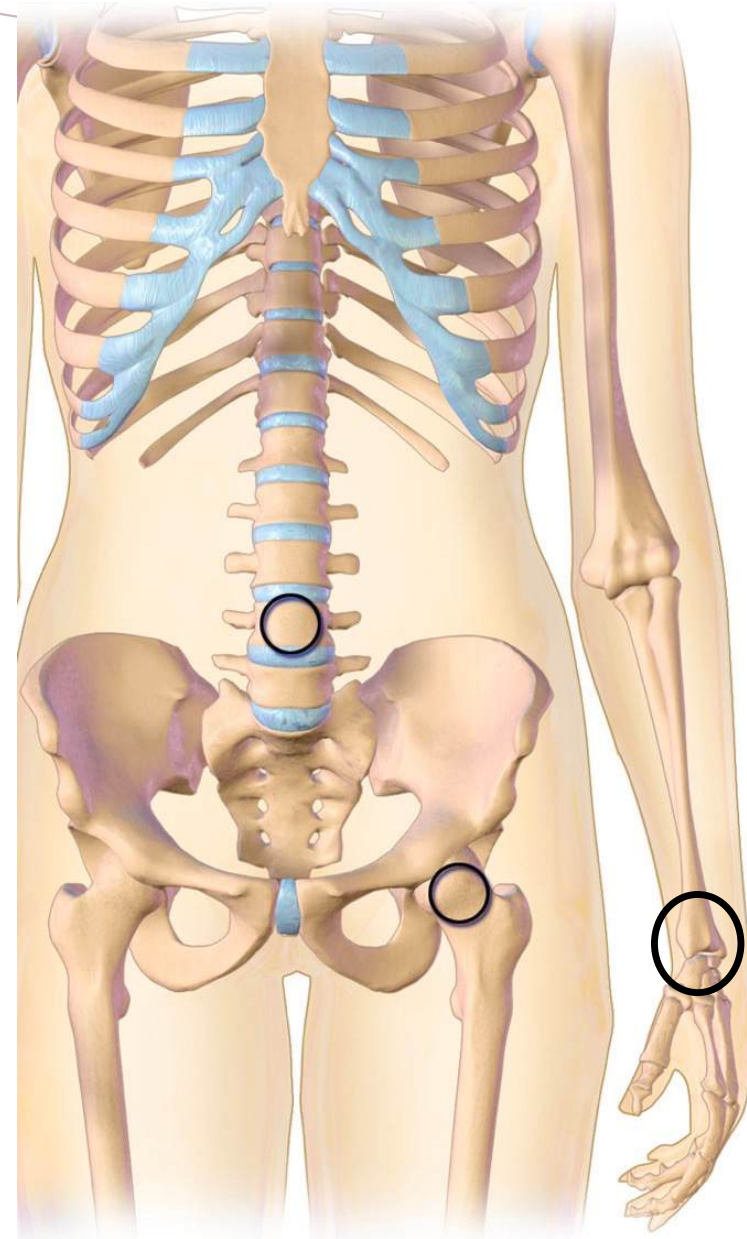
CANCER

- Extra Reading Materials

1. WHO, "Cancer". https://www.who.int/health-topics/cancer#tab=tab_1
2. American Cancer Society, "American Cancer Society Updates Guidelines for Diet and Physical Activity", 2020. <https://www.cancer.org/latest-news/american-cancer-society-updates-guideline-for-diet-and-physical-activity.html>

OSTEOPOROSIS

- A disease characterized by low bone mass and structural deterioration of bone tissue (BMD 2.5 SD or more below the young adult mean)
- Common fracture sites: hip (proximal femur), vertebrae, wrist
- Affects more women than men
- Osteopenia: BMD between 1.0 and 2.5 SD below the mean

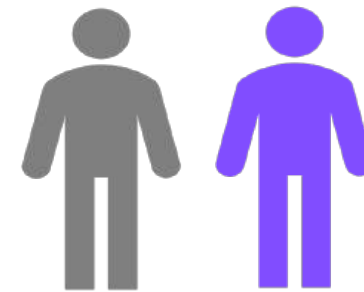


OSTEOPOROSIS

By 2020,
About **1** in **2** Americans >50-
year-old will have osteoporosis

BONE BREAKING INCIDENCE, >50-year-old

1 in **4**   **1** in **2**



Risk Factors

Reproductive
Hormones

Inadequate Ca &
Vit D

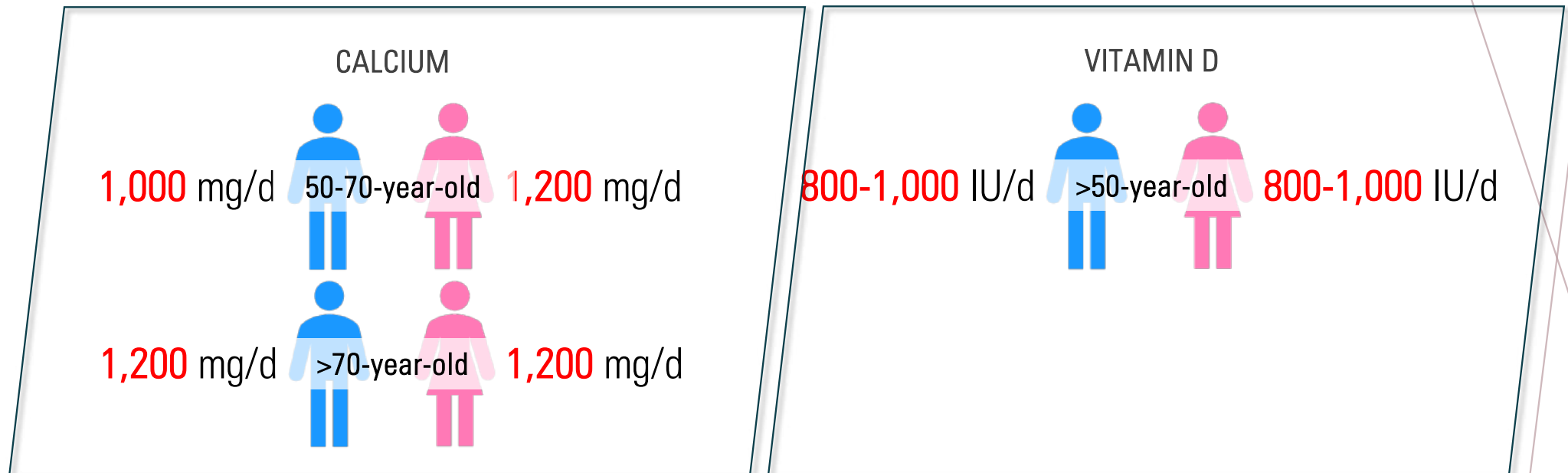
Physical inactivity

Genetics

Female,
Postmenopausal

Caucasian/Asian

OSTEOPOROSIS – LIFESTYLE



- Excessive alcohol intake impairs bone health
- Smoking impairs acquisition of peak bone mass in adolescents and young adults

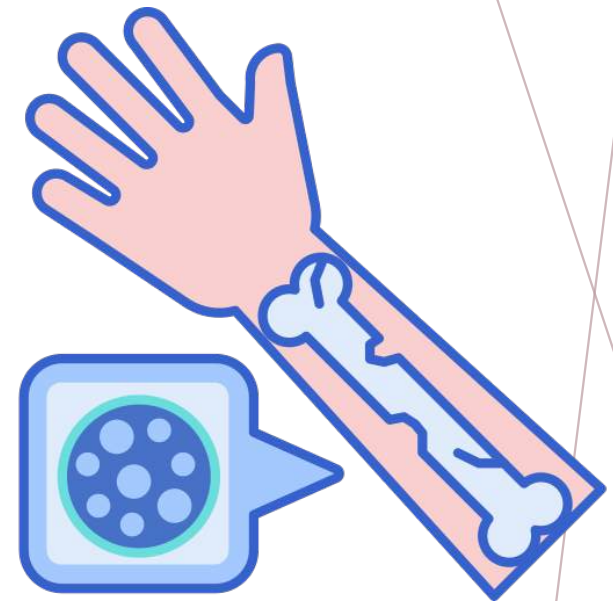
OSTEOPOROSIS – EXERCISE

Goals of physical exercise:

- Prevent bone loss
- Decrease fall risk

Considerations:

- Muscular training, weight-bearing and impact exercises can effectively stimulate bone growth
- Incorporate balance training: improve strength in the trunk, hips, and lower extremities
- Avoid forward flexion, twisting, and compression of the spine
- Avoid high fall-risk movements (quick, jarring movements), slick surfaces, and tripping hazards



OSTEOPOROSIS – EXERCISE

	Frequency	Intensity	Time	Type	Progression
CT	4-5 d/wk	<VT1 40-60% HRR Some may tolerate higher intensity	20 min ↓ 30 min ↓ 40-60 min max.	Weight-bearing E.g. walking, stair climbing, dancing, aerobics, jogging/running	Follow ACE-IFT Model
RT	1-2 d/wk ↓ 2-3 d/wk	High, at least 2 reps are challenging to perform	≤8-10 exrs/session 1x8-12 ↓ 2x8-12	All major muscle groups Emphasize bone-loading Standing exr emphasizing balance, gait, functional movements	Follow ACE-IFT Model

Table 13-15: Exercise Guidelines Summary for Clients with Osteoporosis

OSTEOPOROSIS

- Extra Reading Materials:

1. National Osteoporosis Foundation, "Osteoporosis Exercise for Strong Bones".
<https://www.nof.org/patients/treatment/exercisesafe-movement/osteoporosis-exercise-for-strong-bones/>
2. NIH, "Osteoporosis". <https://www.bones.nih.gov/health-info/bone/osteoporosis/overview>

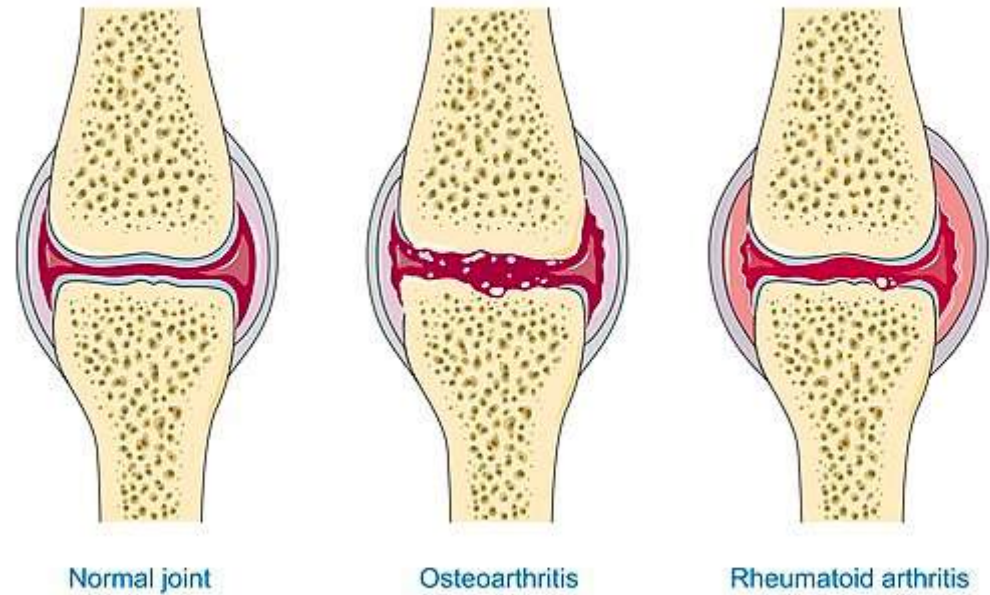
ARTHRITIS

- A chronic condition that causes joint pain

Most Common Types:

- Osteoarthritis (OA)
- Rheumatoid arthritis (RA)

Osteoarthritis and rheumatoid arthritis



ARTHRITIS – RA

- An autoimmune disease that causes chronic inflammatory disorder in the joint
- The most crippling form of arthritis
- Causes great joint deformity, typically in the fingers, knuckles wrists, and feet
- Symptoms: joint pain, swelling, stiffness, contractures (i.e. Permanent tightening of tissues)



ARTHRITIS – OA

- A degenerative arthritis that leads to deterioration of cartilage, resulting in wear and tear of the bone and development of bony spurs.
- Most common form of arthritis
- Causes: overuse, trauma, obesity, degeneration of joint cartilage
- Common sites: cervical & lumbar spine, hip, knee, proximal & distal interphalangeal joints
- Symptoms: pain, swelling, stiffness after periods of inactivity, loss of flexibility

Normal Knee

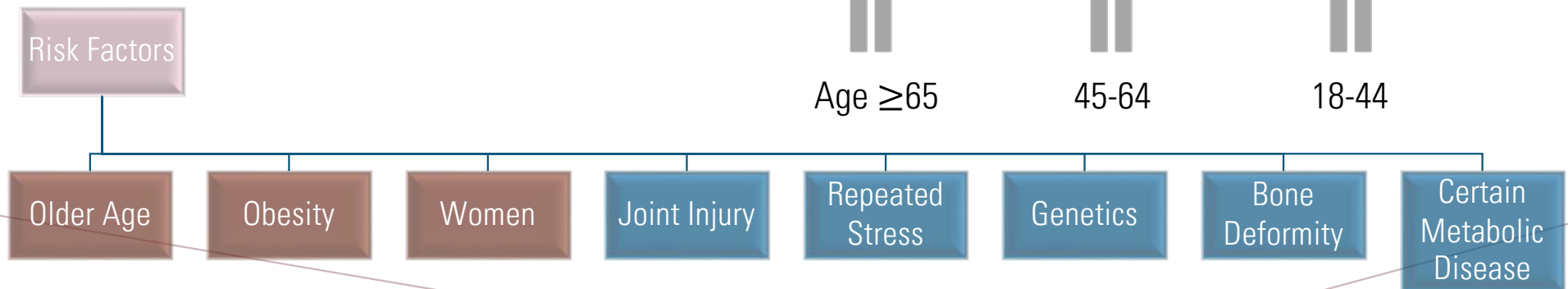
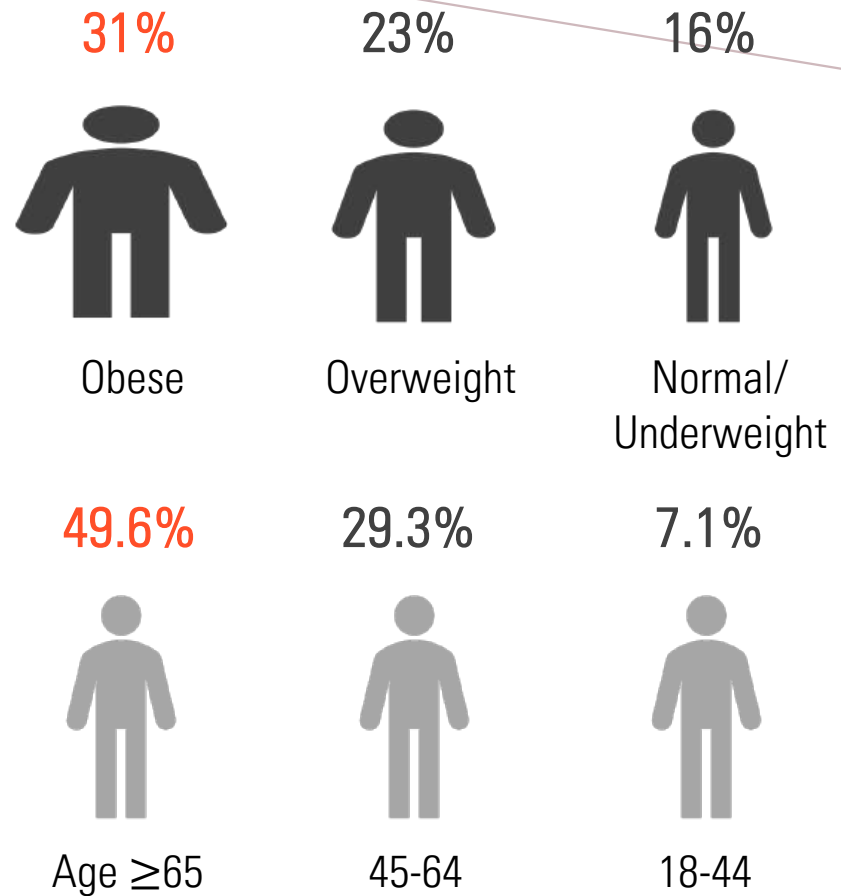
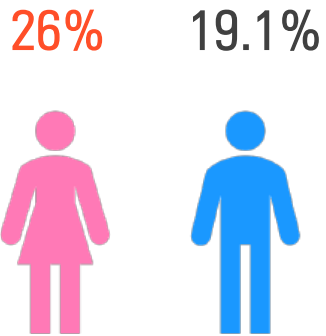


Osteoarthritis



Source from: Heathline

ARTHRITIS – OA



ARTHRITIS – EXERCISE

- Goals:
 - Improve disease condition, improve cardiorespiratory fitness, increase muscular endurance & strength, improve ROM, restore physical functions, reduce CVD risk (for RA)
 - Educate the clients about the importance of regular exercise
- Program Considerations:
 - Generally same as exercise recommendations for healthy adults
 - Consider current level of activity, pain, and functional limitations
 - RA: low-impact exercises using limited ROM, exercise in warm water, avoid exercise during flare-ups

ARTHRITIS – EXERCISE GUIDELINES

	Frequency	Intensity	Time	Type	Progression
CT	3-5 d/wk	Light: 30-40% HRR for deconditioned clients Moderate: <VT1, 40-60% HRR, or Vigorous: ≥60% HRR	≥150 min/wk light/moderate or ≥75 min/wk vigorous or combination	Various, e.g. walking, cycling, swimming <u>Not recommended</u> for lower-extremity arthritis: high-impact activities, e.g. running	Follow ACE-IFT Model
RT	2-3 d/wk	50-80% 1RM Start low	2-4x8-12	All major muscle groups Machines, FW, BW Pain-free ROM	Follow ACE-IFT Model

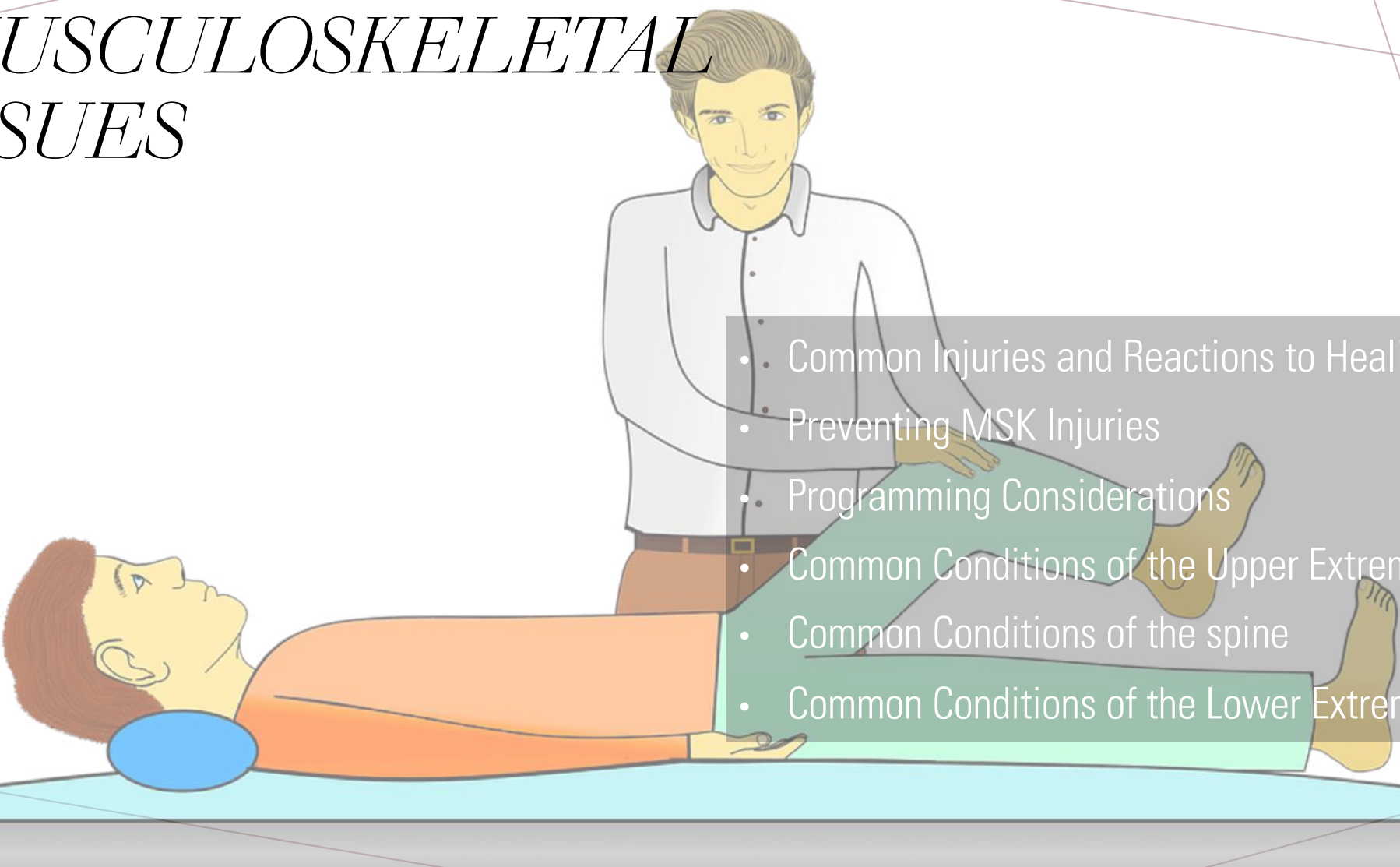
Table 13-17: Exercise Guidelines Summary for Clients with Arthritis

SUMMARY – CHRONIC DISEASE



CHAPTER 15
CONSIDERATIONS FOR CLIENTS
WITH MUSCULOSKELETAL ISSUES

MUSCULOSKELETAL ISSUES

- 
- A cartoon illustration of a physical therapist, a man with brown hair wearing a light blue button-down shirt and brown pants, standing and examining the leg of a female patient. The patient is lying on her back on a light blue mat, wearing an orange long-sleeved shirt and green pants. A blue circular cushion is under her head. The therapist's hands are on the patient's right leg, specifically the knee area.
- Common Injuries and Reactions to Healing
 - Preventing MSK Injuries
 - Programming Considerations
 - Common Conditions of the Upper Extremity
 - Common Conditions of the spine
 - Common Conditions of the Lower Extremity

COMMON INJURIES

- **Acute injury**
 - Injuries that occur in an instant
 - Identifiable onset
 - *E.g. popping sensation and sudden pain when tearing a ligament*
- **Chronic injury**
 - Injuries that manifest progressively over time
 - Hard to pinpoint the onset
 - *E.g. slowly developed low-back pain that persists for months; osteoarthritis that often takes several years to develop*

REACTIONS TO HEALING



MOVEMENT RECOMMENDATION		
• Non-weight bearing • Active ROM	• Weight bearing as tolerated • Gentle, pain-free ROM	• Progressive, pain-free exercise
CONTRAINDICATION: Movements causing pain or swelling at site of injury		

COMMON ACUTE INJURIES - STRAINS

- Muscle strains
 - Microscopic tears of the muscle fibers causing swelling, discoloration (ecchymosis), or loss of function.
 - Grade I: Mild pain or intense local soreness/cramping. Some swelling may be present. Muscle strength remains normal
 - Grade II: Moderate. More severe pain and swelling that likely cause weakness and decreased ROM.
 - Grade III: Severe. Complete tear of the muscle. Sudden tear or “pop” with immediate pain and loss of function. Swelling, discoloration may occur. May have noticeable deformation. More common in older population.
- What are the most common joints associated with this type of injury?

COMMON ACUTE INJURIES - SPRAINS

- Ligament sprains
 - A joint injury that results in stretching or tearing of the stabilizing connecting tissues, causing immediate pain, swelling, instability, decreased ROM, and a loss of function
 - Sprains occur most often with trauma:
 - Contact
 - Non-contact
- What are the most common joints associated with this type of injury?

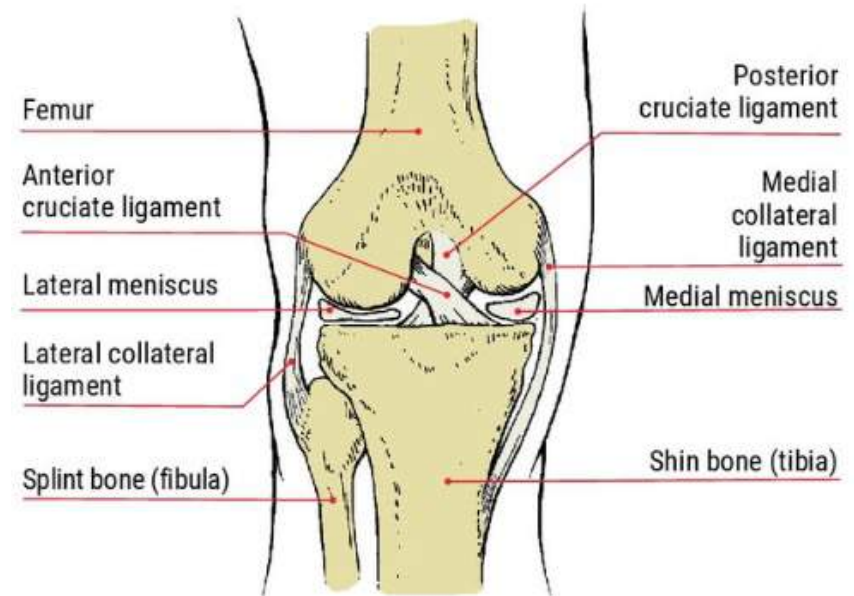


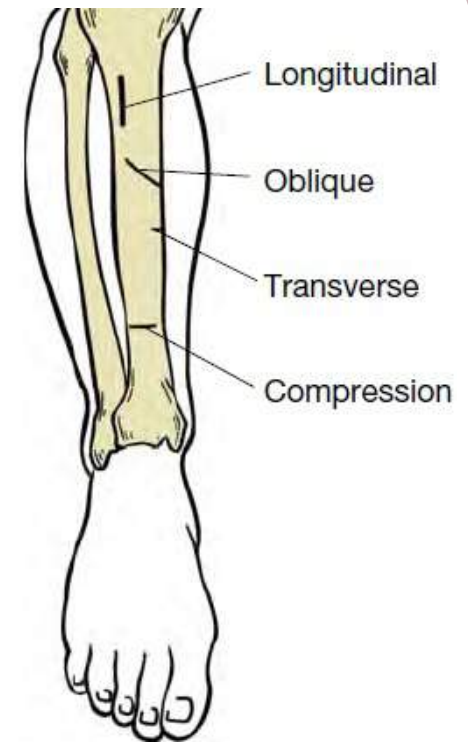
Figure 15-2: Knee Joint Anatomy

COMMON ACUTE INJURIES – CARTILAGE DAMAGE

- Cartilage: collagenous structure that provides shock absorption, stability, joint congruency, lubrication, and proprioception
- Mechanism of injury: compression + sheering
- Common injury site: knee (e.g. meniscal tear)
- Signs and symptoms of meniscal tears:
 - Joint pain and swelling
 - Muscle weakness
 - Stiffness
 - Giving way, locking, clicking or popping

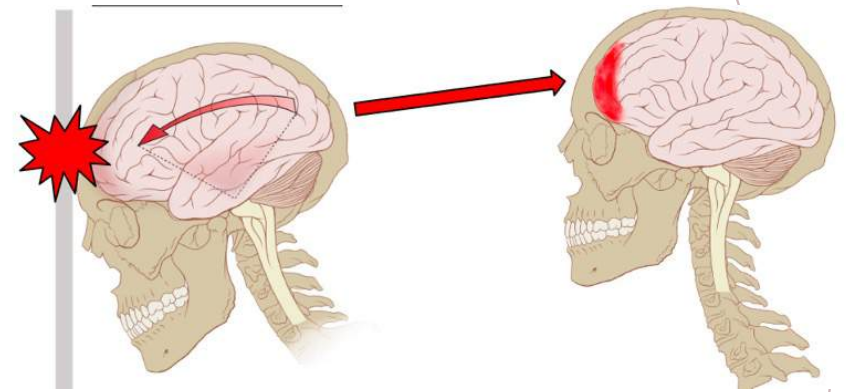
COMMON ACUTE INJURIES - FRACTURES

- Impact fracture:
 - A short fall on a level surface
 - High force impact from accidents or sports
- Pathological fracture
 - Caused by infection, cancer, osteoporosis, or other medical conditions that weaken bones
- Immediate Care:
 - Prevent further injury, provide first aid for shock
 - EMS



COMMON ACUTE INJURIES – HEAD, NECK, BACK INJURIES

- Concussion
 - A brain injury that causes a change in mental status
 - Signs:
 - Confusion, disorientation, memory loss, altered speech, discoordination, impaired balance, dizziness, syncope, less alert, seizure, headache, nausea, or vomiting
 - Second-impact-syndrome:
 - A second injury to head following the first concussion that can be life-threatening
 - Medical care:
 - Stop activity until permission is given by qualified healthcare professional
 - Written return-to-activity protocol
 - Gradual return to activity and close watch for reemergence of signs and symptoms



Source from Max Andrews / CC BY-SA
(<https://creativecommons.org/licenses/by-sa/3.0>)

COMMON ACUTE INJURIES – HEAD, NECK, BACK INJURIES

Acute neck and back injuries

- Strain
- Disc injury
 - Mechanism: compression + sheer
 - Outcome: prolapse of disc tissue
 - Symptom: referred pain/shooting pain

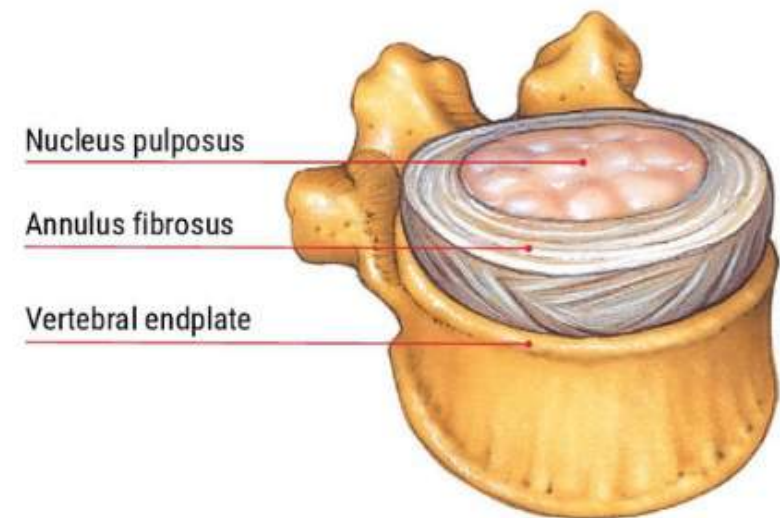


Figure 15-3: Anatomy of a vertebral disc

COMMON OVERUSE CONDITIONS

- Tendinitis (or tendinopathy): inflammation of the tendon
 - Commonly occurs in the shoulders, elbows, knees, and ankles
- Bursitis: inflammation of the bursa sac
 - Commonly affects the shoulders, hips, and knees
- Fasciitis: inflammation of the connective tissue called fascia:
 - Commonly occurs in the bottom and back of the foot
- Stress fracture:
 - Repeated microtrauma to a bone region resulting in a stress fracture

PREVENTING MSK INJURY

- Program design
 - Periodization
 - Client consideration: age, injury history, condition, disease
- Flexibility and elasticity
 - Muscle and postural imbalances, misuse and overuse
- Warm-up
- Recovery

PROGRAMMING CONSIDERATIONS

- Investigation: discover injuries and conditions by using *lifestyle and health-history questionnaire* (Refer to Figure 5-4: pg 150-153)
- Ask questions to gather more information with regard to injury history, pain, activity, and treatment (pg 695-696)
- Program design
 - Periodization
 - Client consideration: age, injury history, condition, disease
- Warm-up, cool-down
- Flexibility, mobility
 - Correct muscle and postural imbalances, prevent misuse and overuse
- Recovery
 - E.g. stretching, self-myofascial release

PROGRAMMING CONSIDERATIONS

- Establish goals
- Use pain scale

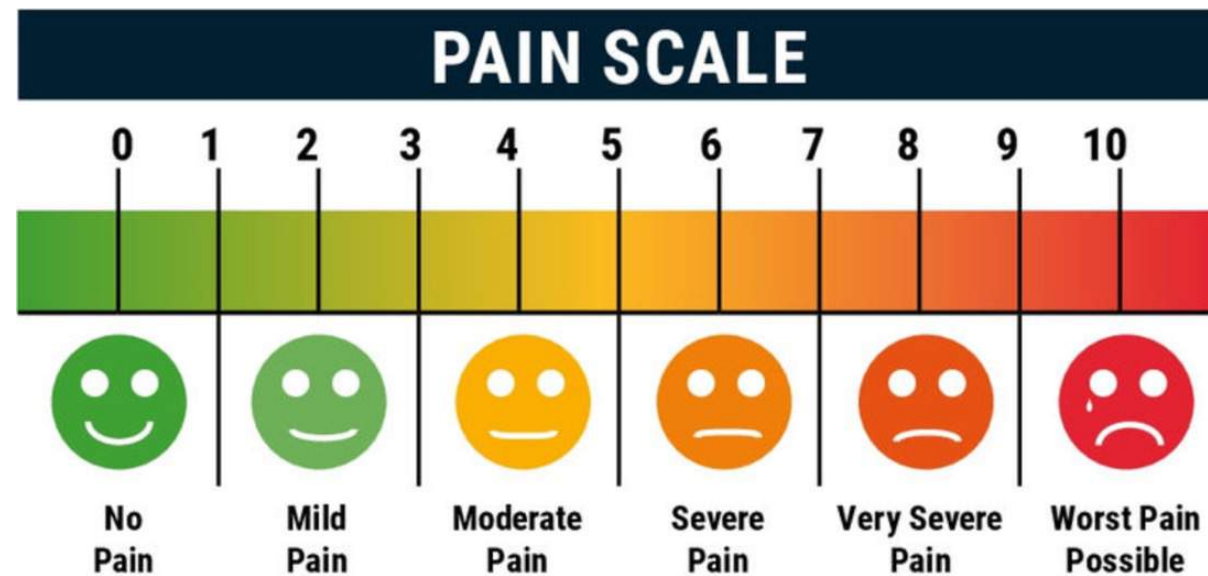


Figure 15-4: Perceived pain scale

PROGRESSION CONSIDERATIONS – ACUTE INJURIES

Type	RT	CT	Avoid
Lower-extremity injury: Affects the kinetic chain in all weight-bearing movements. Upper-body strength training Water-based exercises	• Upper body: seated/lying exercise • Lower body: as prescribed by physical therapist/physician • Spine, core: exercise without affecting injured limb	• Water exercise • Upper-body ergometers • Recumbent/stationary bike	• Any pain-causing movements • Weight-bearing exercise until cleared by physician/physical therapist
Upper-extremity injury: Use common sense	• Lower body: exercise without affecting injured limb • Upper body: as prescribed by physical therapist/physician • Spine, core: exercise without affecting injured limb	• Walking • Recumbent/stationary bike • Elliptical machine	• Any pain-causing movements • Running • Exercise that require both hands for stability
Back/spinal injury:	• As prescribed by physical therapist/physician • BW exercise as tolerated • 8-10 reps	• Water exercise • Walking • Recumbent/stationary bike	• Any pain-causing movements • High reps • Weighted spinal rotation • Getting onto floor

Table 15-3: Muscular and Cardiorespiratory Training Considerations for Post-rehabilitation

SHOULDER

SHOULDER

Joint characteristics

- Ball and socket, shallow glenoid fossa, largest ROM, lack of stability

Common issues

- Impingement of tendons at the subacromial space
- Inflammation of the tendons, bursa
- Rotator cuff muscle tendon tear

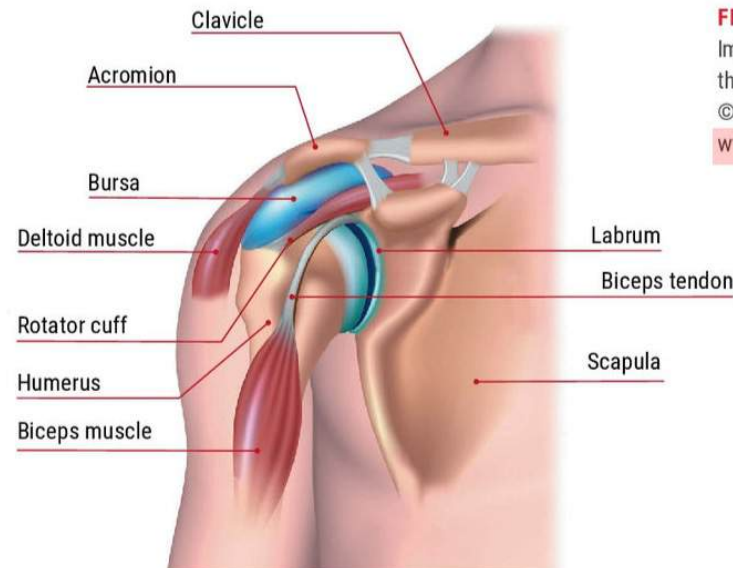


FIGURE 15-6
Impingement syndrome of the biceps tendon
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www.fotosearch.com

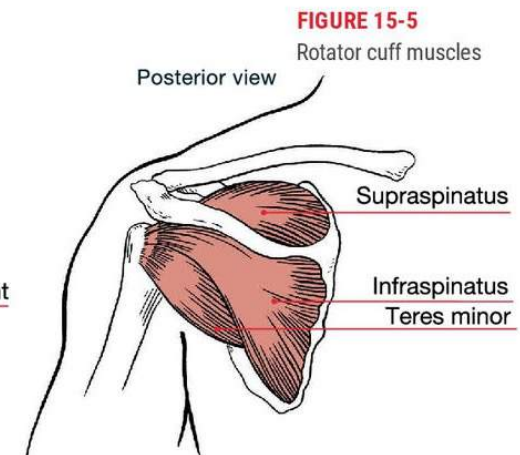
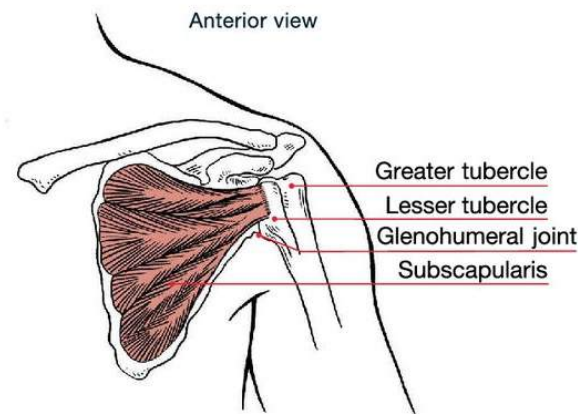


FIGURE 15-5
Rotator cuff muscles

SHOULDER – PROGRAM DESIGN

- Focus:
 - Joint stability, posture correction, strength balance
- Step by step:
 1. Address postural imbalances, open subacromial space: e.g. inhibit/stretch anterior shoulder muscles
 2. Strengthen scapular stabilizers: lower traps, rhomboids, serratus anterior, rotator cuff
 3. Improve strength in anterior shoulder muscles: e.g. controlled pushing
 4. Introduce overhead activities in scapular plane (30° anterior). Modify ROM if needed



Figure 15-8:
Serratus Punch



Figure 15-9:
Farmer Carry



Figure 15-10: Overhead
Press in scapular plane

ELBOW, WRIST, AND HAND

ELBOW, WRIST, AND HAND – ELBOW TENDINITIS

- Lateral epicondylitis (**tennis elbow**)

- An overuse injury of the **wrist extensor** muscle tendons near the origin on the **lateral epicondyle** of the humerus

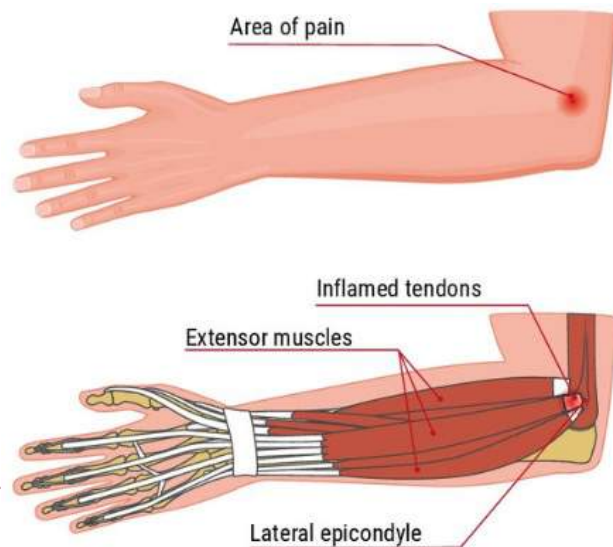


Figure 15-11: Lateral Epicondylitis

- Medial epicondylitis (**golfer's elbow**)

- An overuse injury of the **wrist flexor** muscle tendons near the origin on the **medial epicondyle** of the humerus

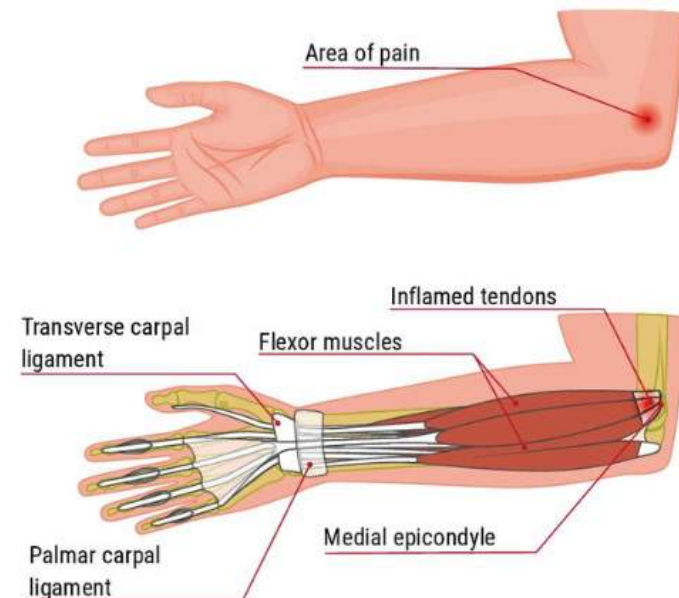


Figure 15-12: Medial Epicondylitis

ELBOW, WRIST, AND HAND – CARPAL TUNNEL SYNDROME

- Caused by pressure on the median nerve
- Symptoms:
 - Pain and numbness
- Recommendation:
 - Ergonomic assessment of workstation
 - Avoid exercises with gripping or wrist extension
 - Use wrist straps
 - Modify to neutral grips

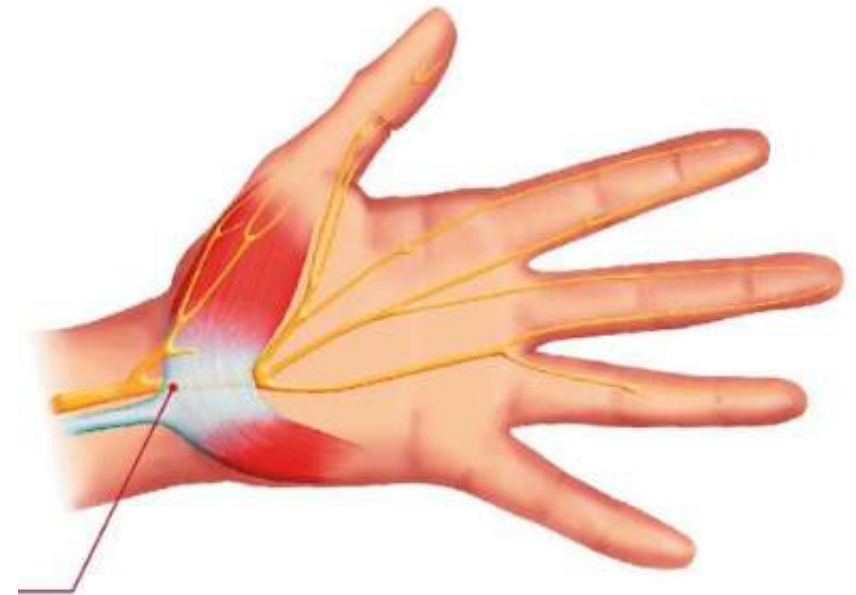


Figure 15-13: Carpal Tunnel Syndrome

ELBOW, WRIST, AND HAND – PROGRAM DESIGN

Limit overuse

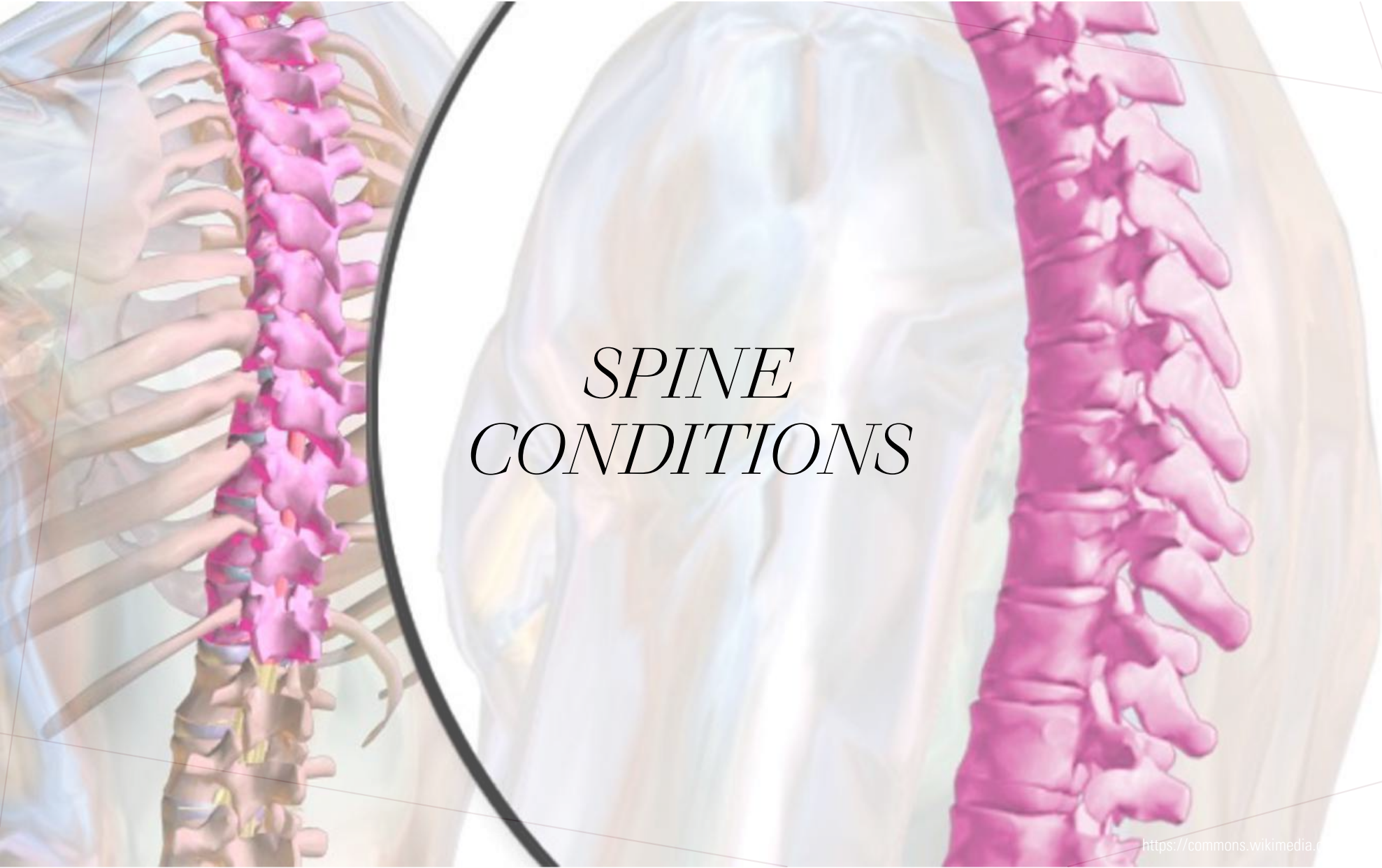
Increase ROM

Improve strength

Remove assistance

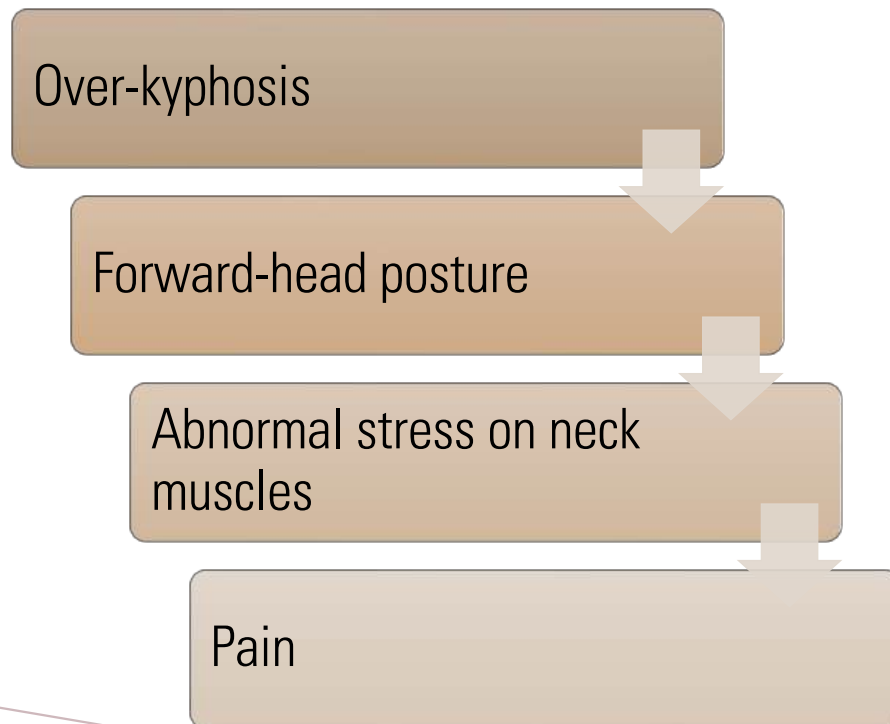
Introduce volume

1. Limit overuse
 - Vary grips, wear gloves
2. Increase ROM through gentle stretches in all planes of motion
3. Improve strength of the wrist and hand
4. Remove grips or gloves, slowly introduce volume to pulling exercises, as tolerated



SPINE CONDITIONS

SPINE CONDITIONS – NECK



Postural alignment

Forward-head position

Figure 15-14: Alignment of the acromioclavicular joint with the ear

SPINE CONDITIONS – NECK

1. Posture assessment of the neck, shoulder and thorax; address the spine and shoulder position
2. Teach the client gentle movement and stretching (as seen here)
3. Refer to a physician if pain persists or worsens



Figure 15-15: Lateral Neck Stretch



Figure 15-16: Rotational Neck Stretch

SPINE CONDITIONS – LOW BACK

Causes:

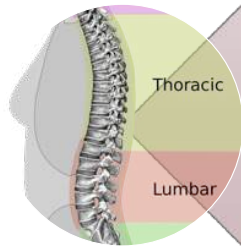
- Muscle strain, soft-tissue stress from prolonged faulty posture, poor physical fitness, disc disease, osteoarthritis, rheumatoid arthritis, ankylosing spondylitis, cancer, etc.

Muscle Group	Function to Support Low-back Pain
Gluteus maximus	Hip extensor; provides support for pelvic and trunk stability; contributes to proper pelvic position
Hamstrings	Hip extensors; provide support for pelvic and trunk stability; contribute to proper pelvic position Tight muscles inhibit hip hinge and may cause posterior tilt.
Rectus abdominis	Trunk stabilizer; contributes to proper pelvic position
Internal and external obliques	Trunk stabilizers; contribute to proper pelvic position
Latissimus dorsi	Provides stabilization of lumbar spine through its attachment at the thoracolumbar fascia

- Stretch overactive muscles
- Strengthen weak muscles
- Improve hip & thoracic ROM

Table 15-4: Select Muscles and Muscle Groups and their contributions to Trunk and Pelvic

SPINE CONDITIONS – PROGRAM DESIGN



Address faulty posture

- Left-right imbalance
- Anterior-posterior imbalance



Address ROM restrictions

- ↗ internal/external rotation of the hip



Strengthen posterior chain muscles

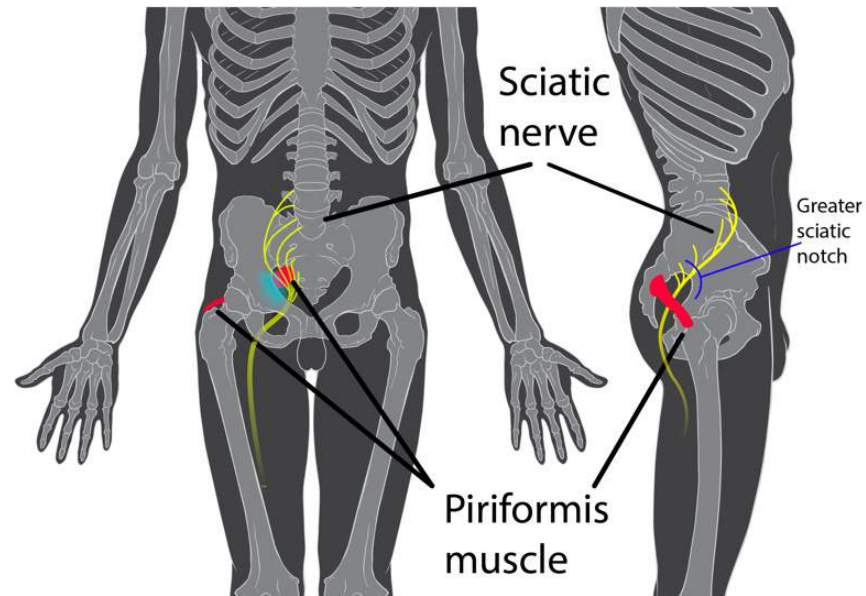
- Gluteal muscles
- Hamstrings

HIP



HIP – PIRIFORMIS SYNDROME

- An uncommon neuromuscular disorder when the piriformis muscle compresses the sciatic nerve
- Symptoms:
 - Pain, tingling, or numbness in the buttocks, may extend down the leg
- Cause:
 - Macrotrauma (majority)
 - Microtrauma from overuse
 - Shortening / elongation of PM resulting in compression / irritation



Source from:

https://commons.wikimedia.org/wiki/File:Piriformis_syndrome.jpg

HIP – PIRIFORMIS SYNDROME

Recommendation:

- Static posture assessment of the hips
- Balance the position of right and left pelvis via static stretching of the tight muscles and conditioning of the taut muscles
- Work on active ROM on both sides
- Improve mobility & stability of the hip through CKC internal / external hip rotation

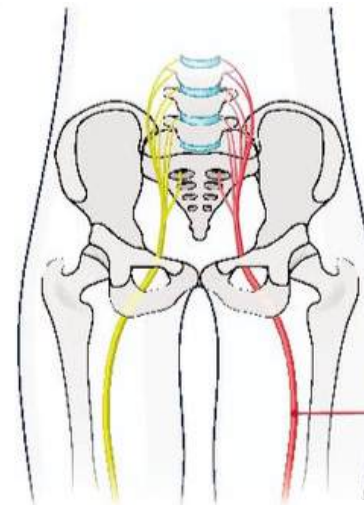


Figure 15-21: Sciatic nerve

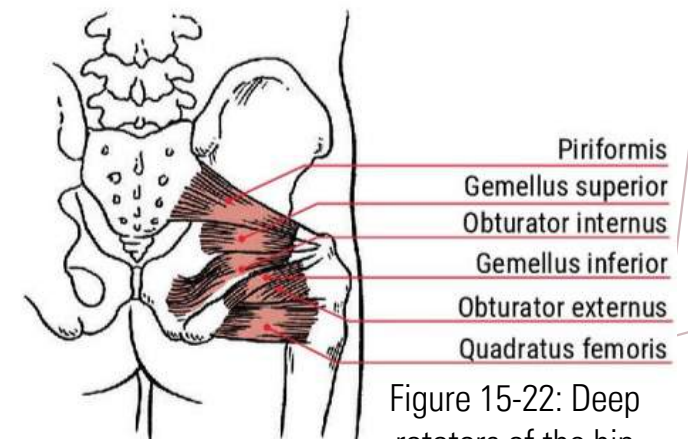
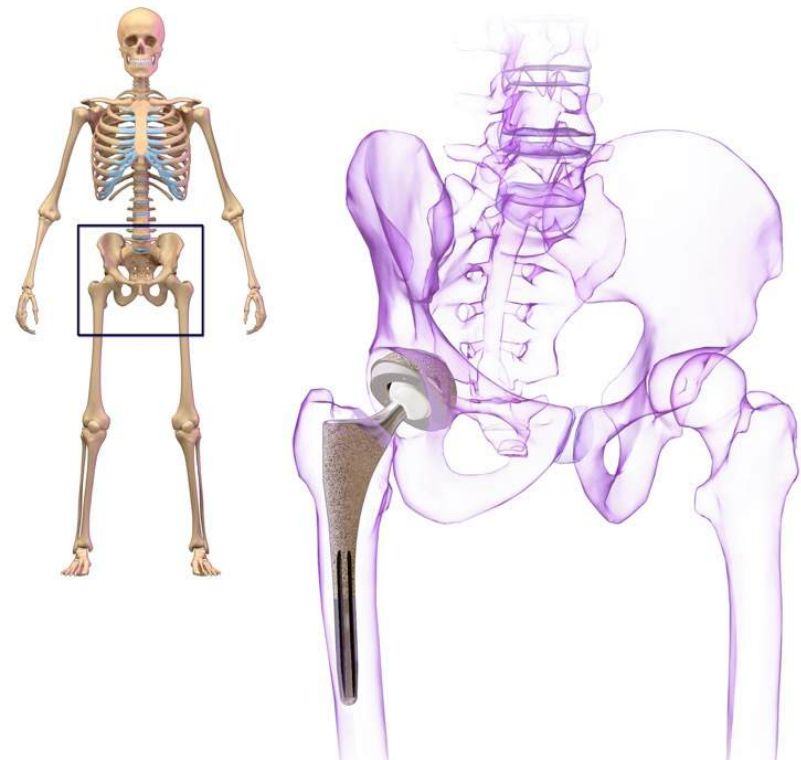


Figure 15-22: Deep rotators of the hip

HIP ARTHRITIS AND HIP REPLACEMENT

- Hip replacement (arthroplasty) is recommended when physical therapy and medication cease to have a positive effect on arthritis and ADL is impacted
- Recommendation:
 - Gentle stretching and dynamic ROM exercises
 - Recumbent bike



Total Hip Replacement

Source from: https://commons.wikimedia.org/wiki/File:Hip_Replacement.png

HIP – PROGRAM DESIGN



Left-right Imbalance



ROM



SL Stability



Hip-hinge stability &
mobility



Strength of posterior chain

1. Address imbalance between left and right sides
2. Improve ROM in hip internal and external rotation
3. Improve stability in single-leg stance
4. Improve hip-hinge mobility and stability
5. Improve strength of posterior chain muscles

KNEE



KNEE – ITBS

- Iliotibial band friction syndrome (ITBS)
 - A repetitive overuse condition when the distal portion of IT band rubs against the lateral femoral epicondyle, causing burning or pain at the lateral aspect of the knee (may be described as sharp or stabbing pain)
 - Common among active individuals 15-50 years old
- Mechanism:
 - Poor training form, lack of recovery, poor techniques
- Risk factors:
 - Overuse, improper footwear/equipment, changes in running surface, muscle imbalance, structural abnormalities (e.g. pes planus, knee valgus, leg-length discrepancy)

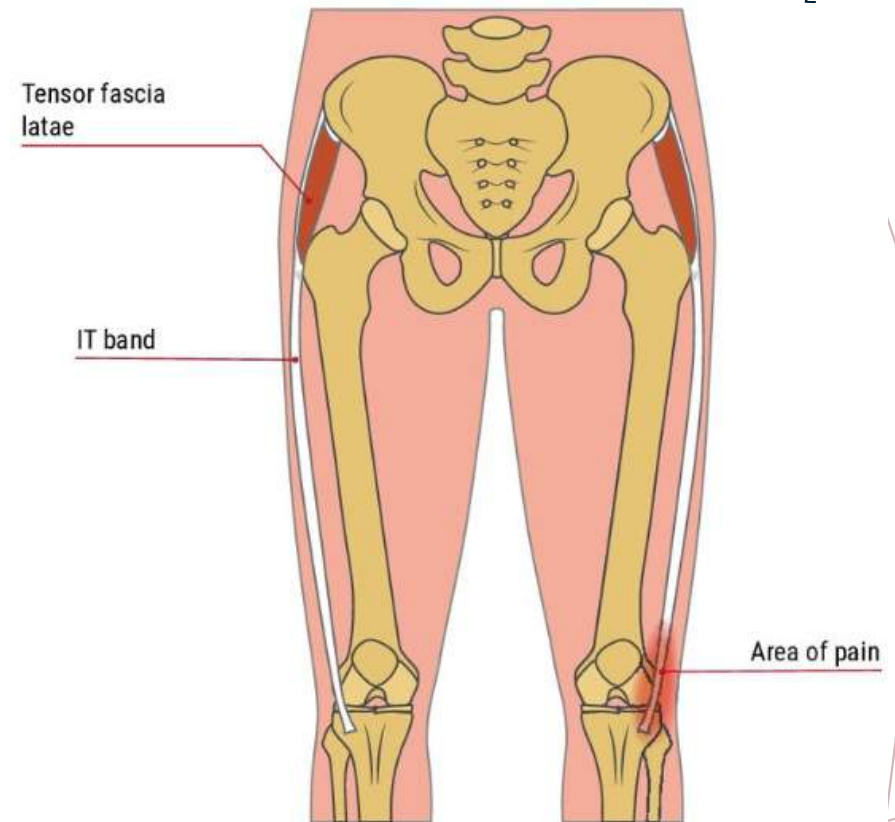


Figure 15-23: Iliotibial band friction syndrome

KNEE – ITBS

Recommendation:

- Posture assessment: to discover positional imbalance of the pelvis
- Self-myofascial release, stretching



Figure 15-24: Self-myofascial release of the iliotibial band



Figure 15-28: Self-myofascial release of the quadriceps



Figure 15-25: Static stretch for the iliotibial band

KNEE – TENDINITIS

- Inflammation of tendons causing localized pain
- Common in running, stair-climbing, squatting
- Common sites:
 - Patellar tendon
 - Pes anserine tendons
 - Quadriceps tendon

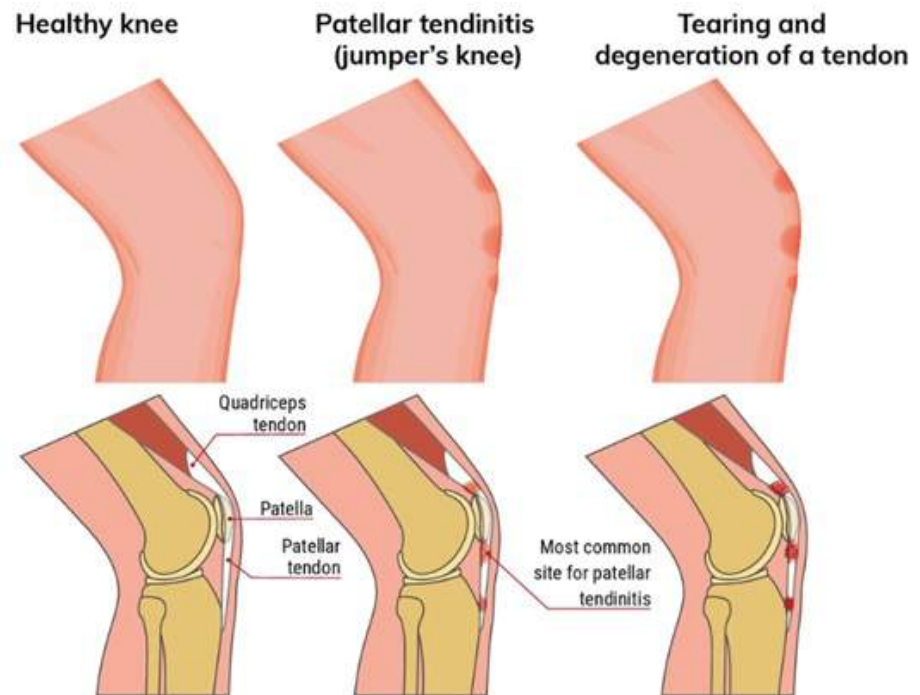


Figure 15-26: Common sites of knee tendinitis

KNEE – TENDINITIS

Recommendation:

- Static posture assessment: foot pronation/supination
- Movement assessment: observe hip, ankle, foot
- Identify any foot, ankle, or hip imbalances
 - Typical tightness occurs in the quadriceps

Management:

- Self-myofascial release
- Stretching
- Muscular training to restore strength in the posterior chain

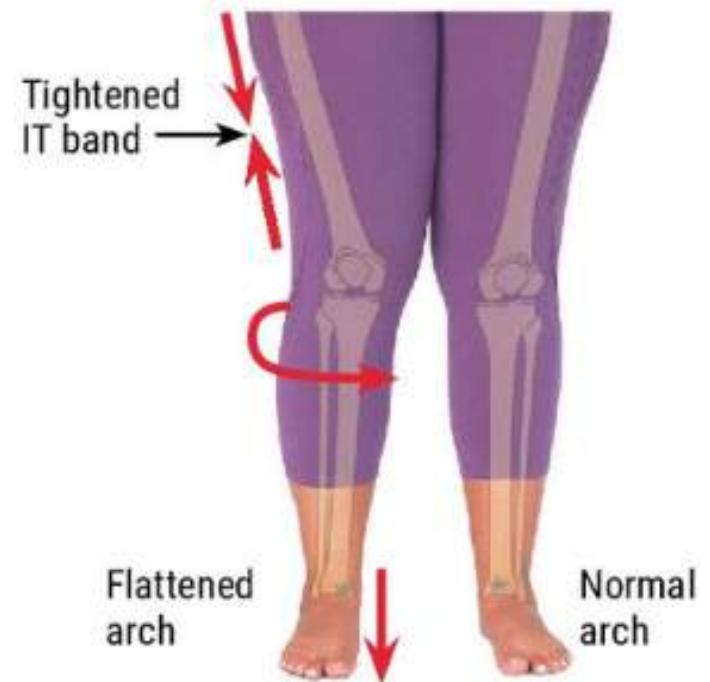


Figure 15-27: Example of foot-position effects on the kinetic chain of the lower extremity

KNEE - PFPS

Patellofemoral Pain Syndrome (Runner's Knee)

- A diffuse inflammation of the soft tissue surrounding the patella, resulting in generalized patellofemoral pain.

Cause

- Overuse, altered biomechanics, muscle dysfunction
- Improper training technique or equipment
- Recent changes in intensity, frequency, duration
- Changes in footwear or landing surface

Mechanism

- Repetitive loading $\Rightarrow$ abnormal stress to the knee
- Excessive loading exceeds body's physiological balance



KNEE - PFPS

Risk Factor:

- Biomechanical abnormality
 - E.g. pes planus, knee valgus
- Muscle weakness in quadriceps, hip abductors/external rotators
- Muscle tightness and length deficits
- Chondromalacia patella

Recommendation:

- Identify areas of tightness, instability, or postural malalignment
- Muscular strengthening in corresponding hip
- Stretching



Figure 15-31: Bilateral genu valgum, or "knock knees"

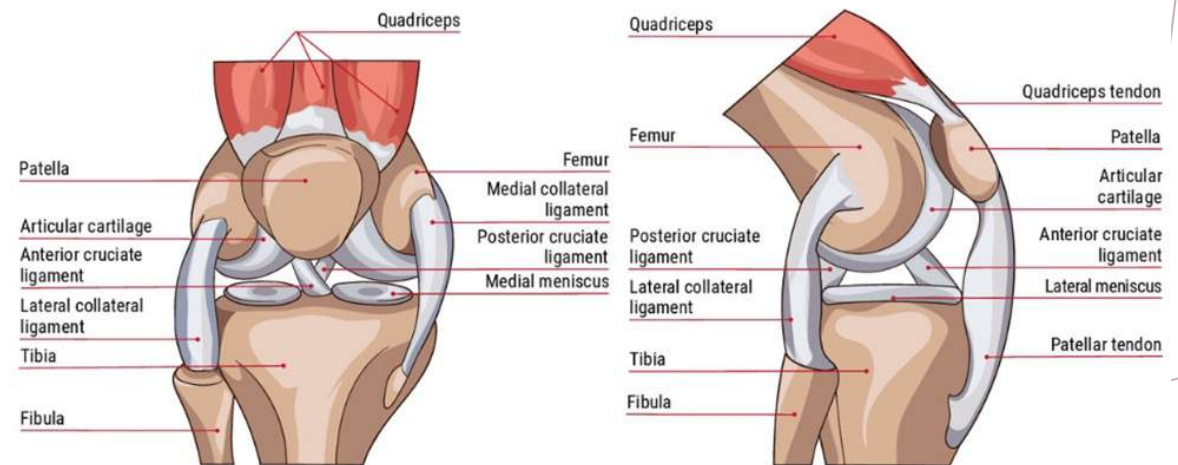


Figure 15-30: Patellofemoral joint

KNEE – KNEE ARTHRITIS AND KNEE REPLACEMENT

- Knee arthritis is a common condition of the knee
- Recommendation:
 - Muscular strength exercise to improve stability, movement efficiency
- Knee replacement/total knee arthroplasty
 - When conservative treatment of knee arthritis has failed



KNEE – PROGRAM DESIGN

1. ROM & movement assessments:
 - Identify mobility, stability and strength deficiency
 - Identify imbalances between left and right sides
 - Observe: hip, ankle, foot alignment
2. Increase ROM, strengthen quadriceps:
 - Stationary/recumbent bike
3. Address any left-right imbalance
4. Address any tightness and instability in the hip, ankle and foot
5. Improve hip hinge mobility and stability
6. Strengthen posterior chain muscles:
 1. Partial ROM squat
 2. Stiff-legged deadlift
 3. Avoid OKC exercises mostly



ANKLE AND LOWER LEG

ANKLE AND LOWER LEG – SHIN SPLINTS

- Exertional, lower-leg pain

Types:

- Posterior shin splints, aka. medial tibial stress syndrome (MTSS)
 - Periostitis.
 - Common in active population: runners, dancers, military personnel
 - Often triggered by sudden change in activity
- Anterior shin splints
 - Common in active population
 - Associated with exertional activity

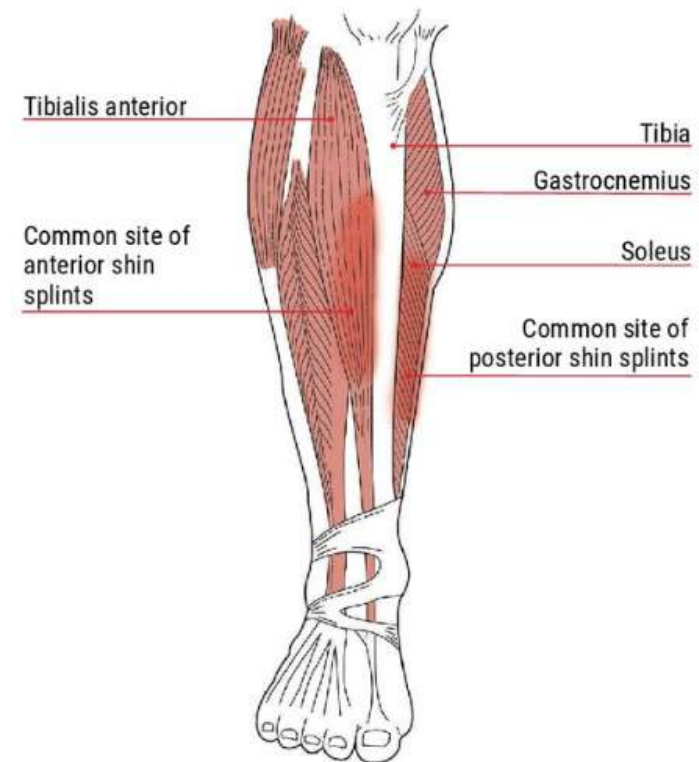


Figure 15-35: Site of pain for anterior and posterior shin splints

ANKLE AND LOWER LEG – SHIN SPLINTS

Program considerations:

- Modified activity
- Cross-training
- Slow return to full activity
- Pain-free stretching of calf muscles, esp. soleus
- Static posture assessment
 - Uncover misalignment or instability in foot/ankle
 - If yes, work on flexibility and stability of the foot and ankle

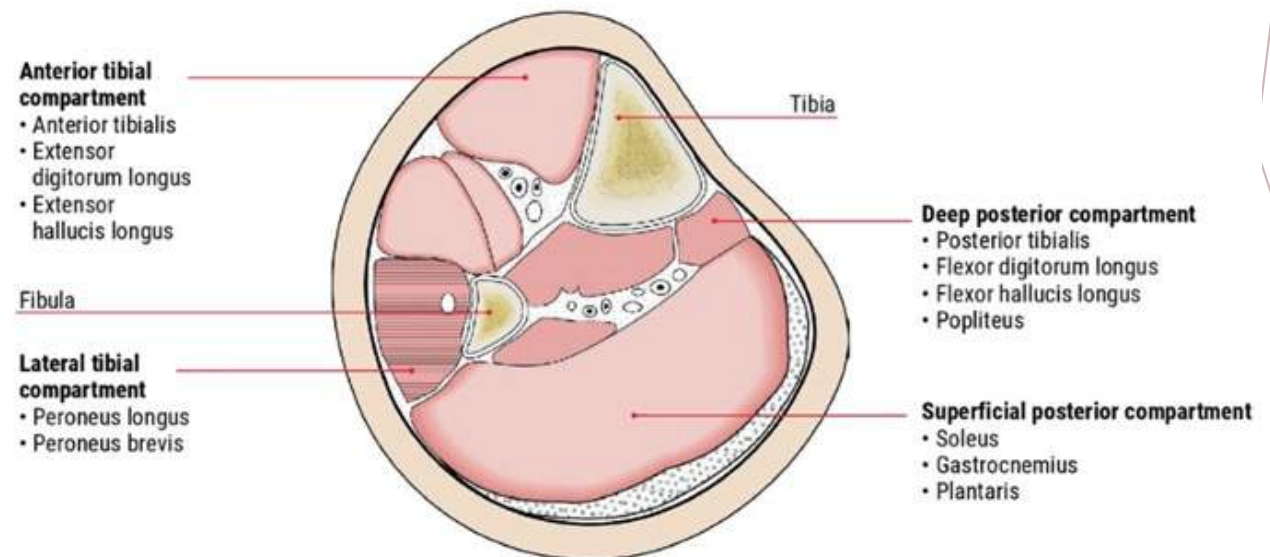


Figure 15-36: Contents of the muscular compartments of the lower leg

ANKLE AND LOWER LEG – ANKLE SPRAINS

- Lateral (inversion) ankle sprain: most common (~75%)
 - Affecting: anterior talofibular ligament, calcaneofibular ligament, posterior talofibular ligament
 - Mechanism of injury: excessive plantar flexion and inversion
- Program focus:
 - Restore proprioception, flexibility, strength
 - Limit exercise on unstable surfaces
 - Use caution with frontal plane exercises to avoid opening the lateral ankle too early

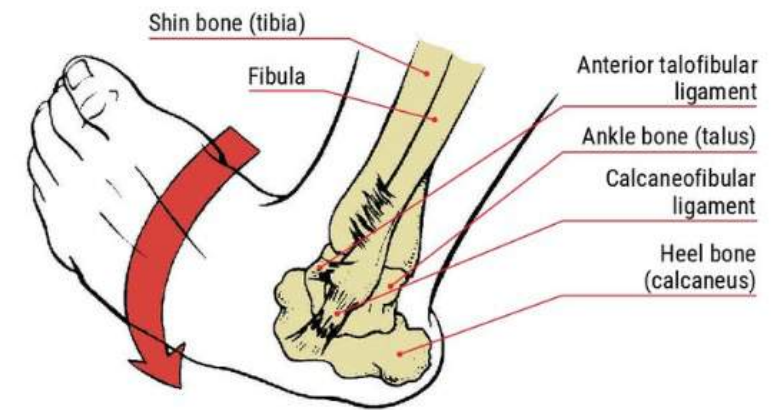
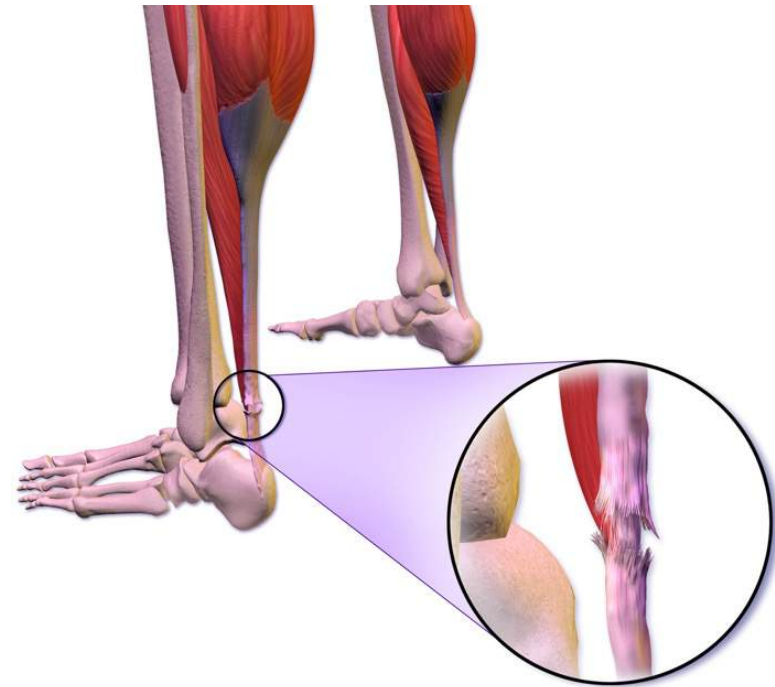


Figure 15-39: Lateral ankle sprain

ANKLE AND LOWER LEG – ACHILLES TENDINITIS

- Inflammation of the Achilles tendon
- Mechanism of injury:
 - Sudden start from a standing position
- Risk factors:
 - Pes cavus (high arch), pes planus (flat feet), leg-length discrepancy, lateral ankle instability
 - Poor training mechanics, poor footwear, muscle weakness, poor flexibility



Achilles Tendon Tear

Source from:
https://commons.wikimedia.org/wiki/File:Achilles_Tendon_Tear.png

ANKLE AND LOWER LEG – ACHILLES TENDINITIS

- Program considerations:
 - Progressively improve elasticity: multiplanar quickness and toe-off propulsion
 - Regain calf flexibility: gentle dynamic stretching recommended



a. Gastrocnemius stretch



b. Soleus stretch

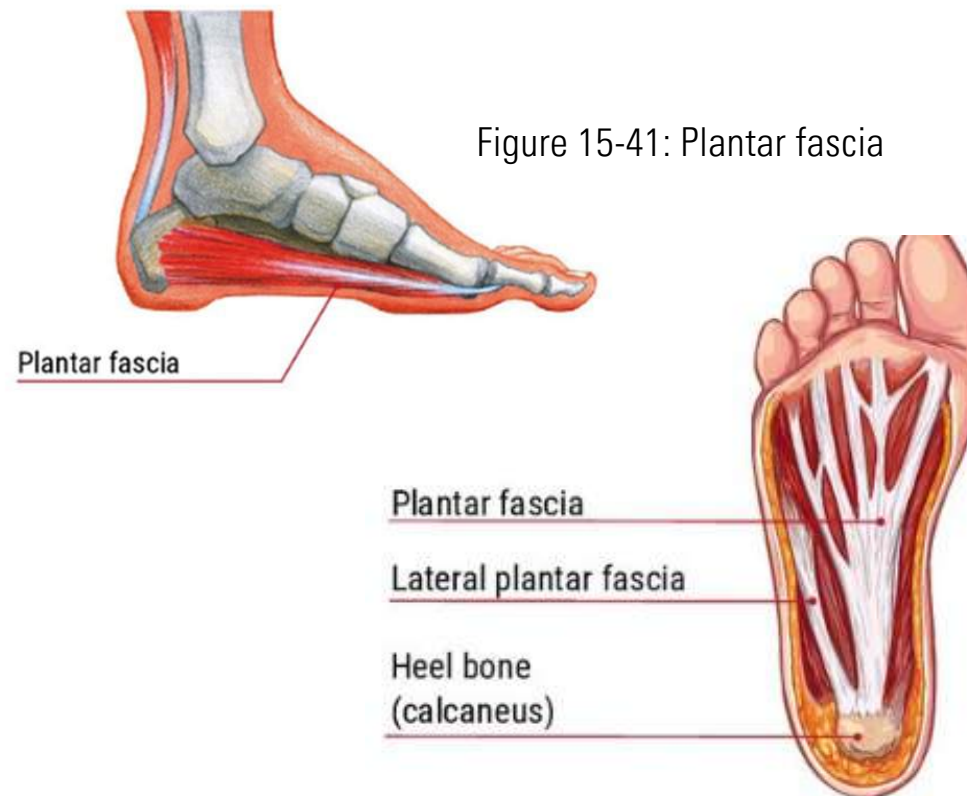
Figure 15-37: Standing calf stretches



Figure 15-38: Calf stretch modifications

ANKLE AND LOWER LEG – PLANTAR FASCIITIS

- Inflammation of the connective tissue plantar fascia
- Indicated exercises:
 - Stretching of calf and plantar fascia
 - Self-myofascial release at the bottom of the foot and calf
 - Strengthen foot muscles to improve arch stability



ANKLE AND LOWER LEG – PROGRAM DESIGN

General considerations:

- Improve function through increasing mobility and stability
- Incorporate balance training

Step by step:

1. Address mobility and stability in the foot and ankle
2. Improve strength in the muscles that control movement in the sagittal plane
3. Improve strength in the muscles that control movement in the transverse plane and frontal plane
4. Improve strength in the posterior chain muscles



COMMON MSK CONDITIONS



- Plantar fasciitis
- Achilles tendinitis
- Shin splints



- IT band syndrome
- Hamstrings strain
- PFPS
- Piriformis syndrome



- Low back pain
- Neck pain



- Shoulder pain
- Tennis elbow
- Golfer's elbow
- Carpal tunnel syndrome

PROGRAMMING FOR A CLIENT WITH PAIN

